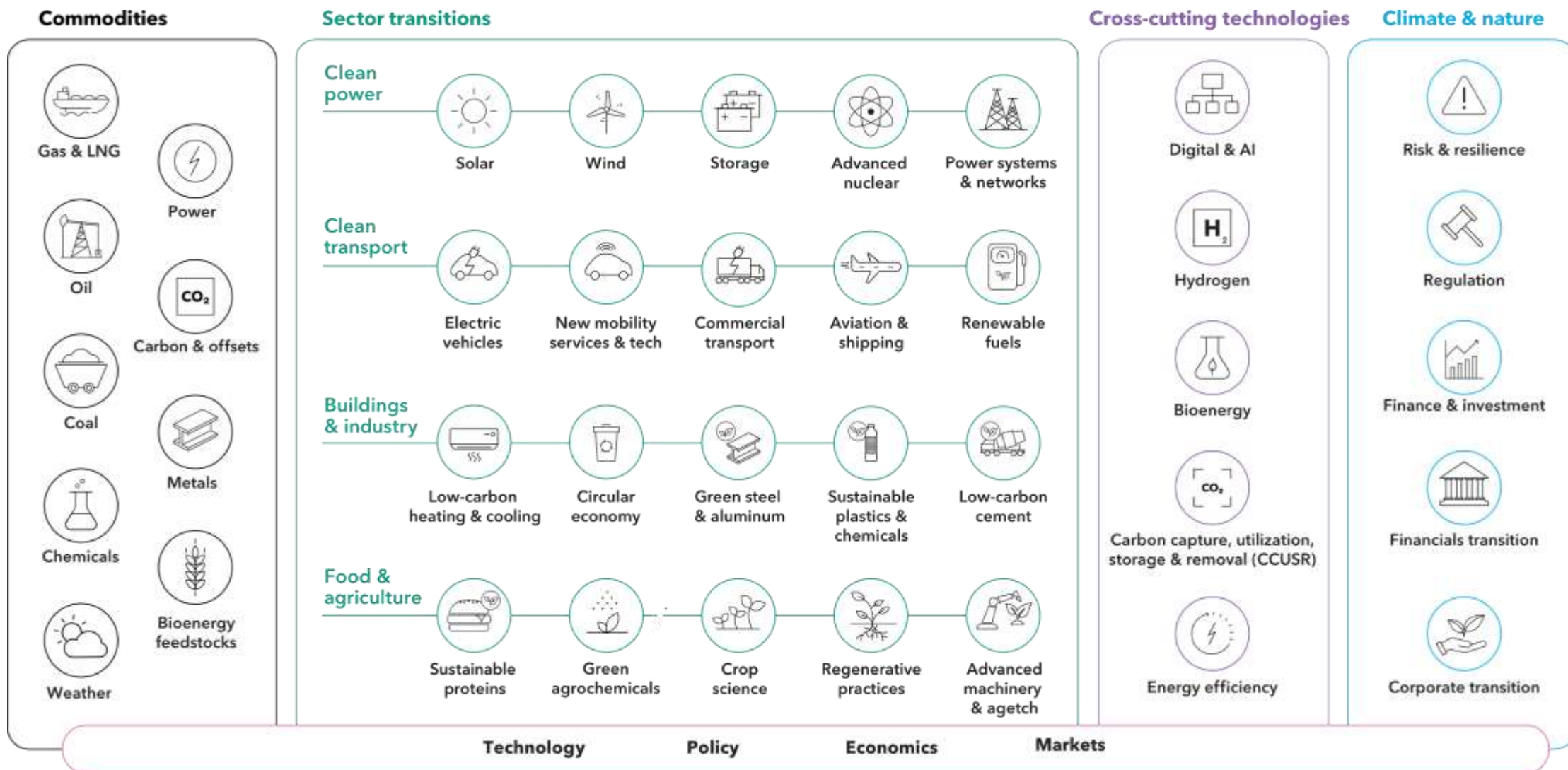


GLOBAL ENERGY MARKETS: CURRENT LANDSCAPE AND REFORM TRENDS

Dr. Nannan Kou

BNEF coverage





And look at the horizon

You have to look at your feet

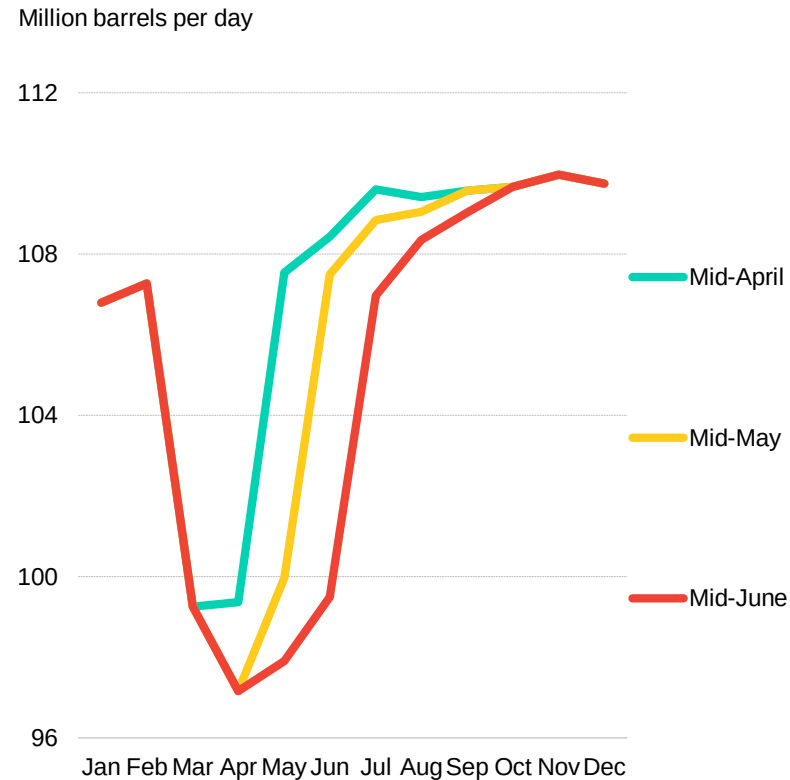
In times of uncertainty...

Image: Cheung 2016

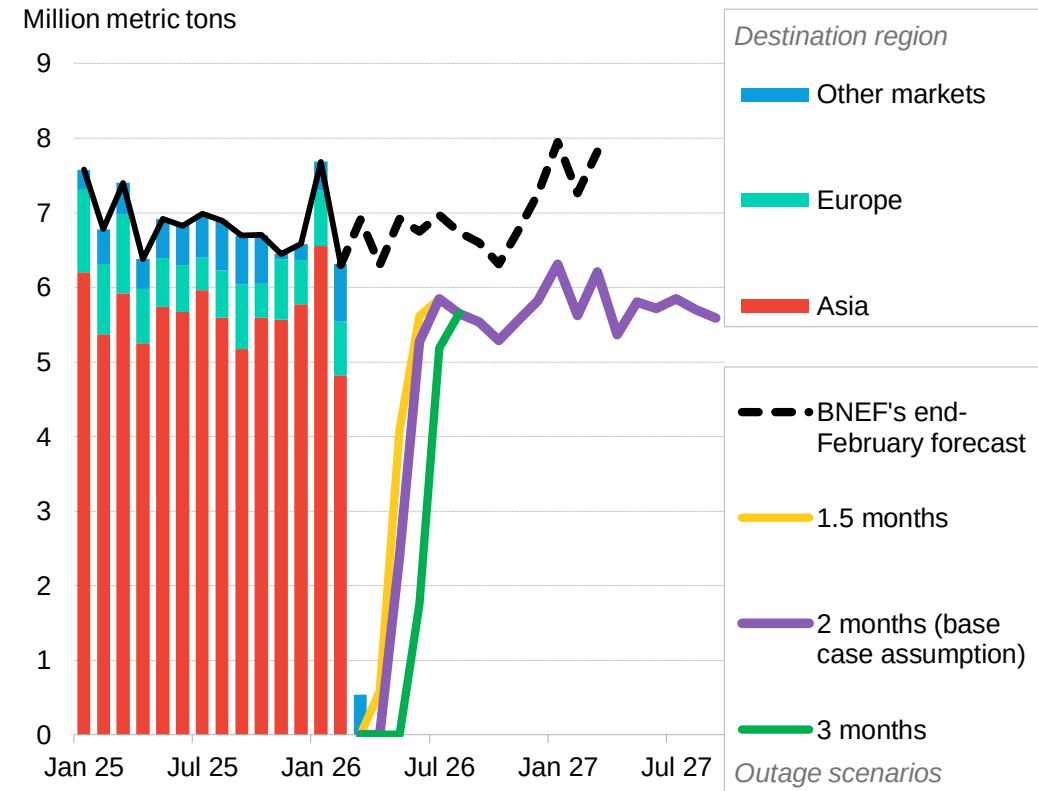


Hormuz crisis: heightened geopolitical risk and enduring uncertainty...

BNEF March oil supply scenarios based on Strait of Hormuz reopening timeline



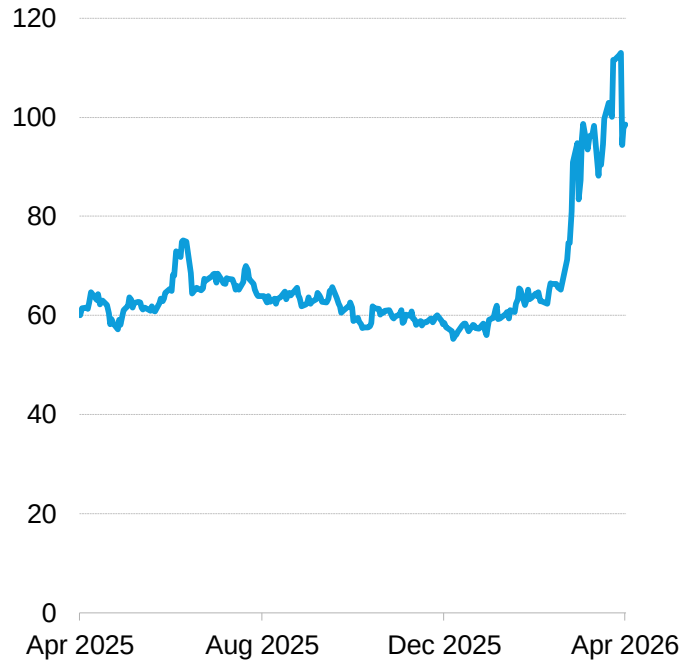
Qatar's LNG exports under three outage scenarios



Source: BloombergNEF, Bloomberg Terminal AHOY JOURNEY <GO>.

...with real economic consequences

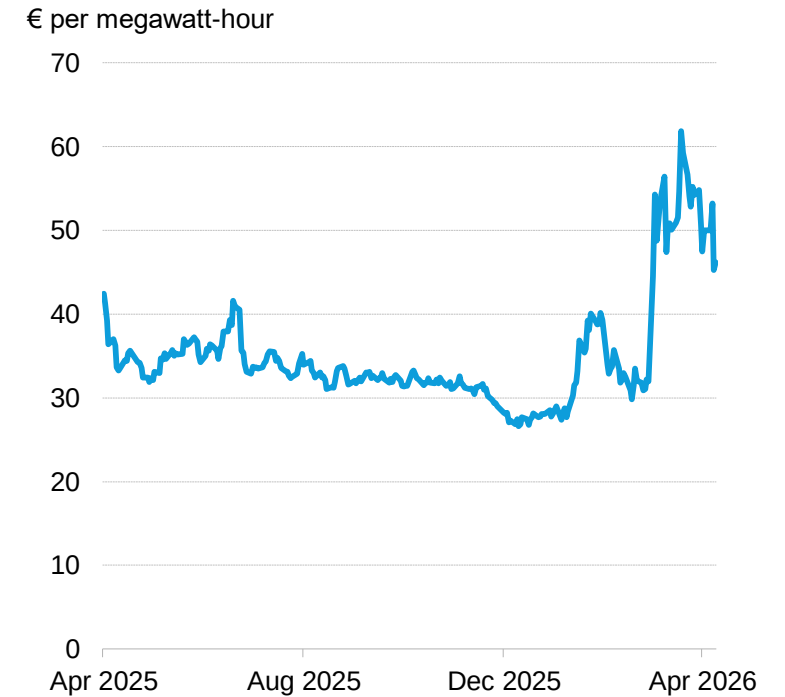
WTI crude, trailing 12 months



JKM LNG, trailing 12 months



TTF natural gas, trailing 12 months

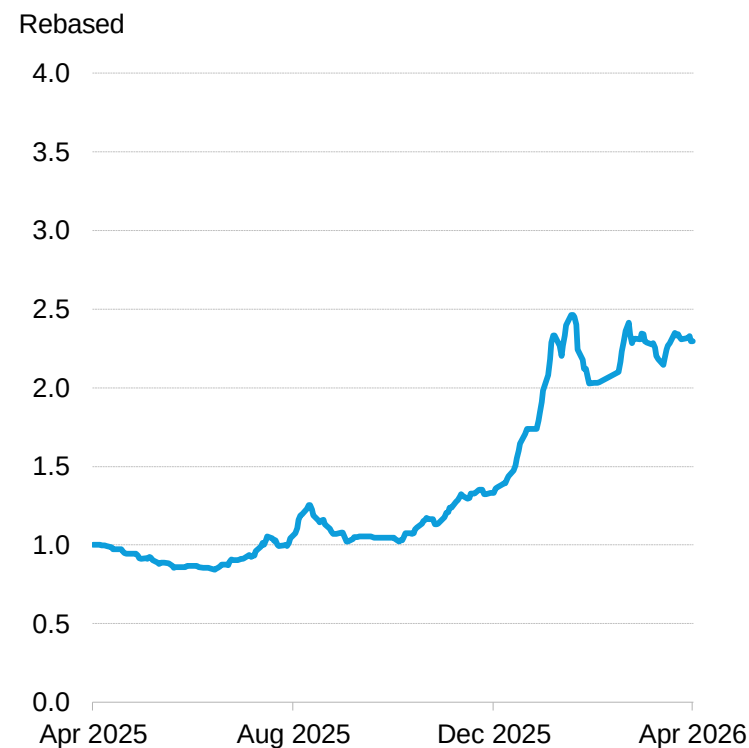


Source: Bloomberg Terminal. Note: Data up to April 10. WTI refers to West Texas Intermediate. JKM refers to the Japan Korea Marker for liquefied natural gas. TTF refers to the Title Transfer Facility. MMBtu is million British thermal units.

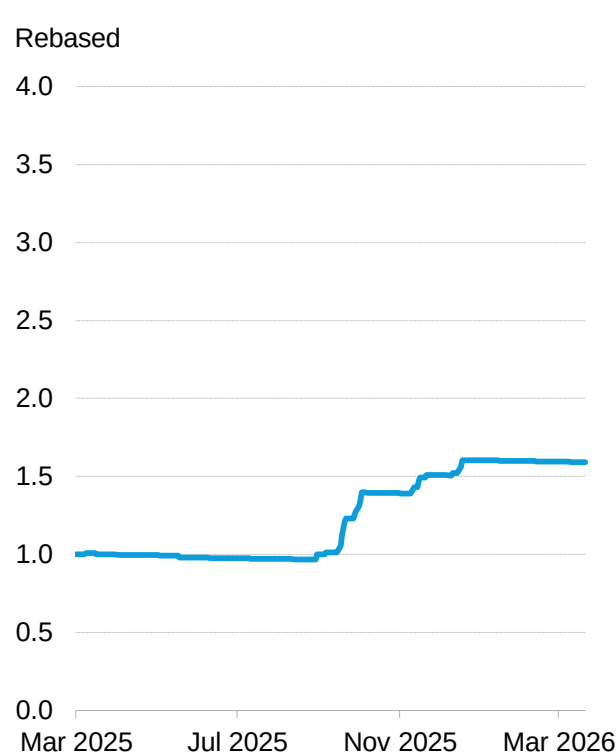


Critical mineral prices are also sensitive to macro risk, affecting equipment prices

Lithium carbonate, trailing 12 months



Cobalt, trailing 12 months



Silver, trailing 12 months

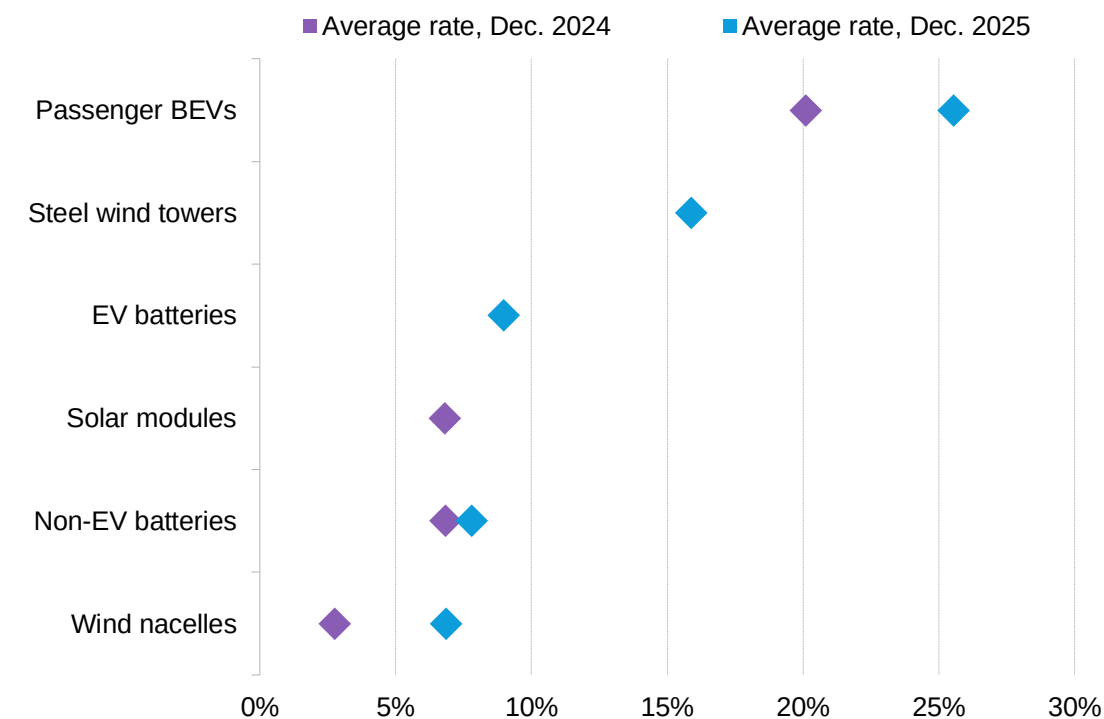


Source: Asian Metals, London Metals Exchange, BloombergNEF, Bloomberg Terminal. Note: Data up to April 10.



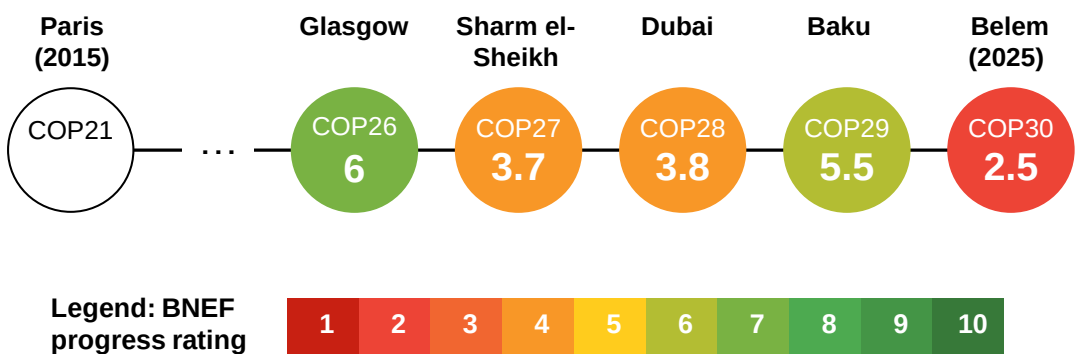
Global cooperation on trade and climate is not working as it used to

Average import tariffs on clean technologies across selected G-20 markets



Source: BloombergNEF. Note: Data shows the simple average tariff applied by the 15 assessed G-20 markets. BEV stands for battery-electric vehicle, and EV is electric vehicle.

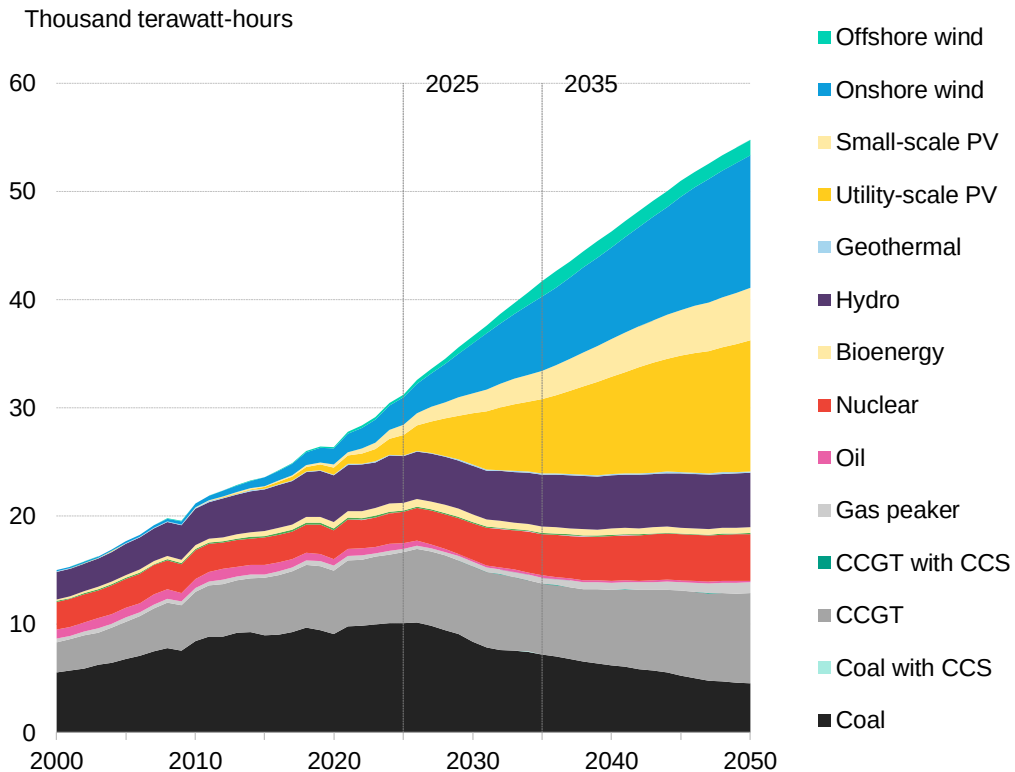
COP achievement scores from Glasgow to Belem



Source: BloombergNEF. Note: Scores are out of 10 for achievement against BNEF's expectations.

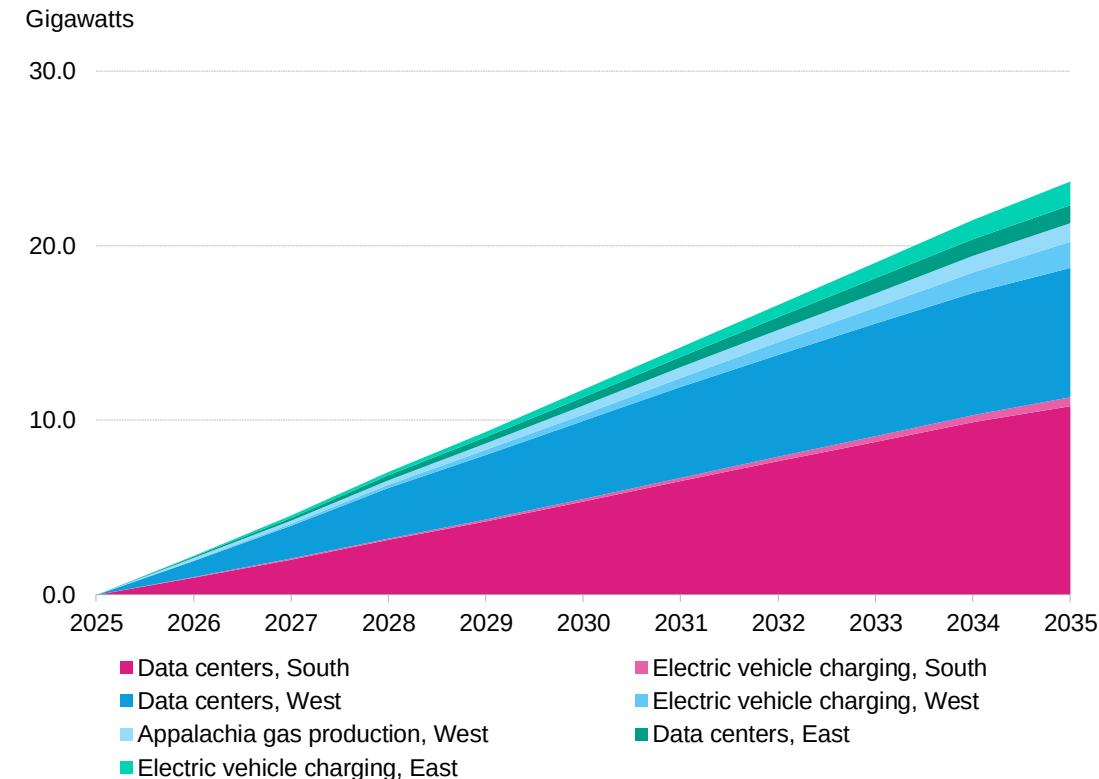
Supply, demand and the external environment are changing at the same time

Electricity generation by technology, Economic Transition Scenario



Source: BNEF

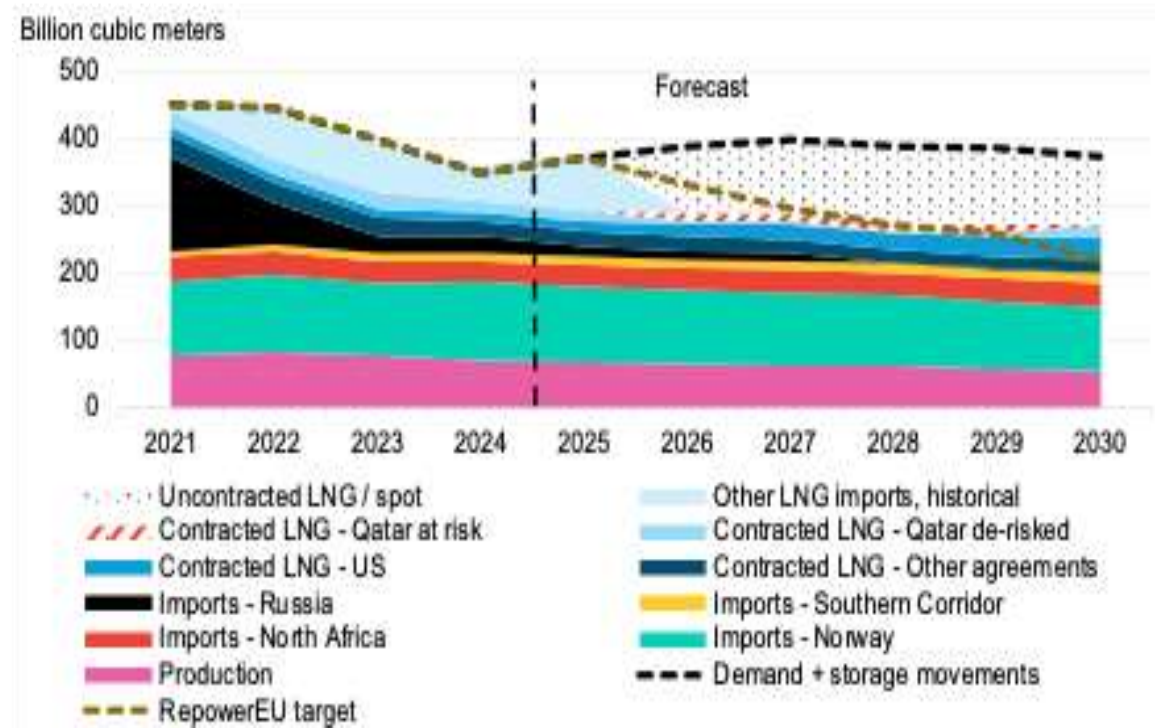
PJM subregional new average hourly load growth



Source: BloombergNEF, DCByte. Note: Labeling for 'West' subregion in the chart corrected on April 6, 2026.

The central question is not just how much power we have, but how we value it

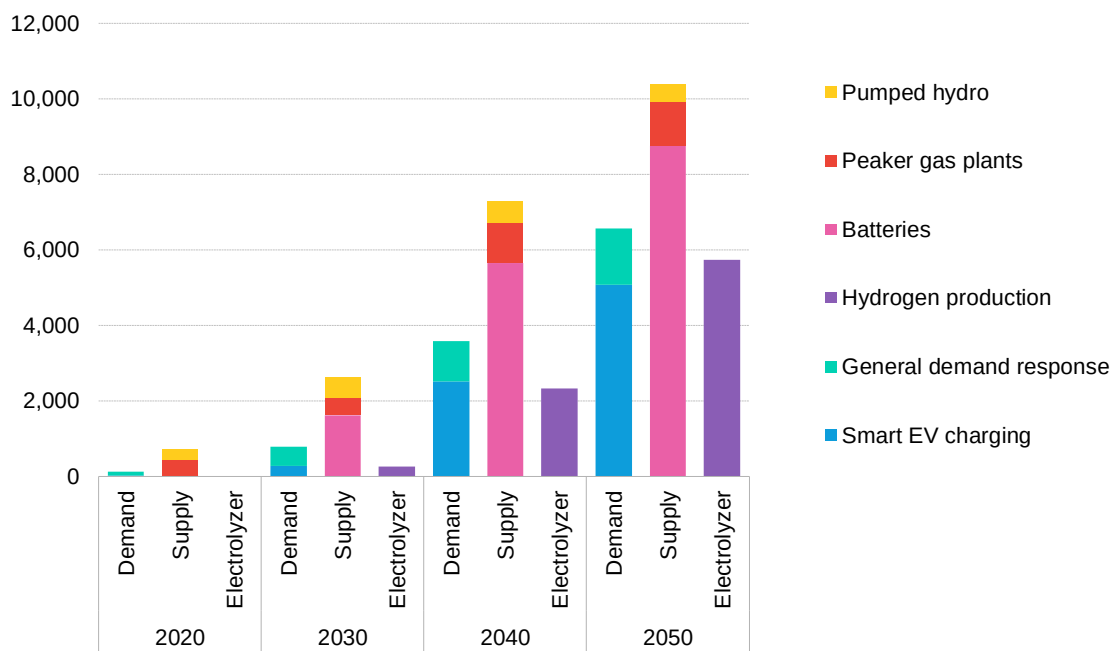
European historical gas supply by source and origin, future contracted LNG volumes, and gap to meet BNEF’s forecasted gas demand



Source: BloombergNEF

Power system flexibility in the Net Zero Scenario

Terawatt-hours of flexibility provided



Source: BloombergNEF

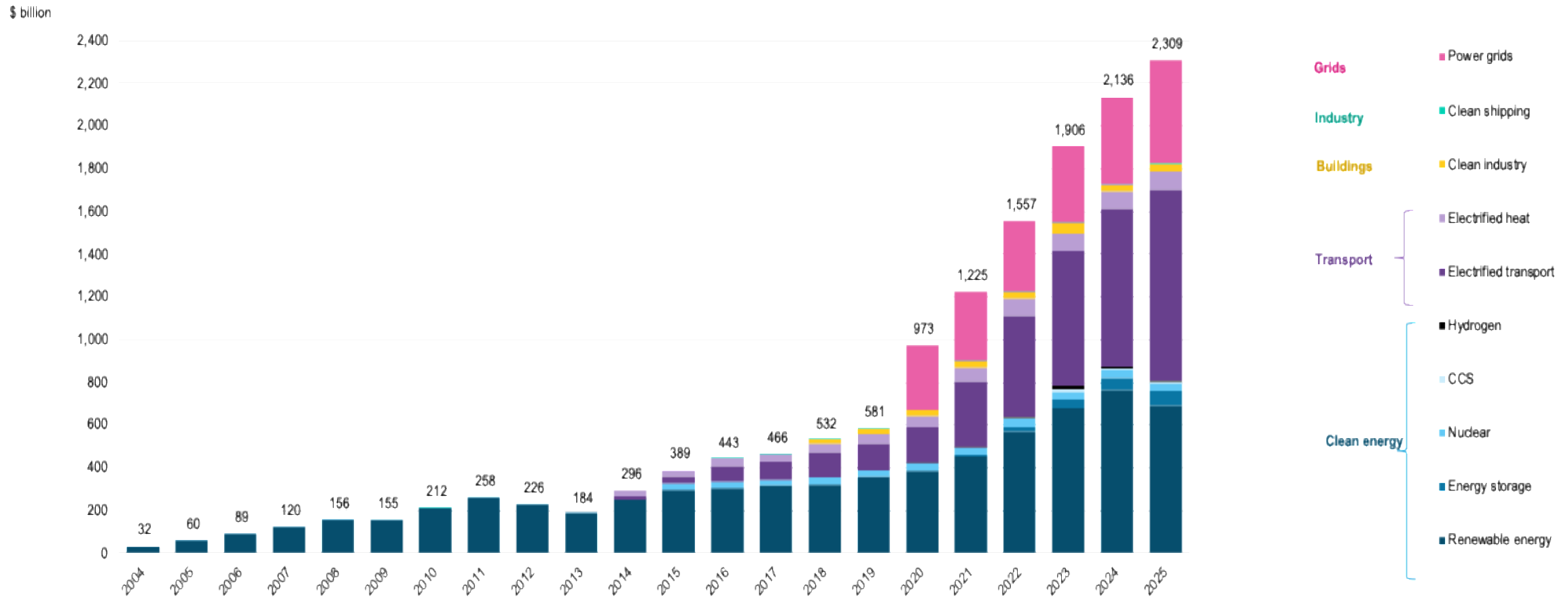
ONE

NEW VARIABLES RESHAPING GLOBAL ENERGY MARKETS



Energy security has re-emerged as a binding constraint on the transition

Global energy transition investment by sector

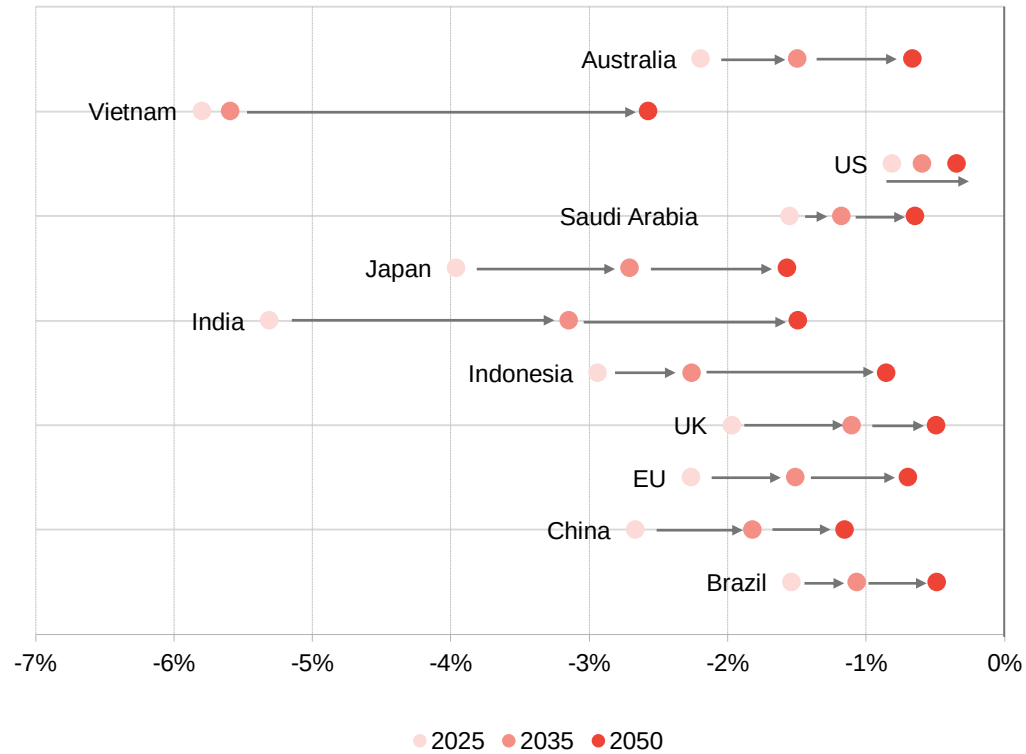


Source: BloombergNEF

Energy security has re-emerged as a binding constraint on the transition

Energy commodity import dependence by market and scenario, 2025, 2035, 2050, ETS

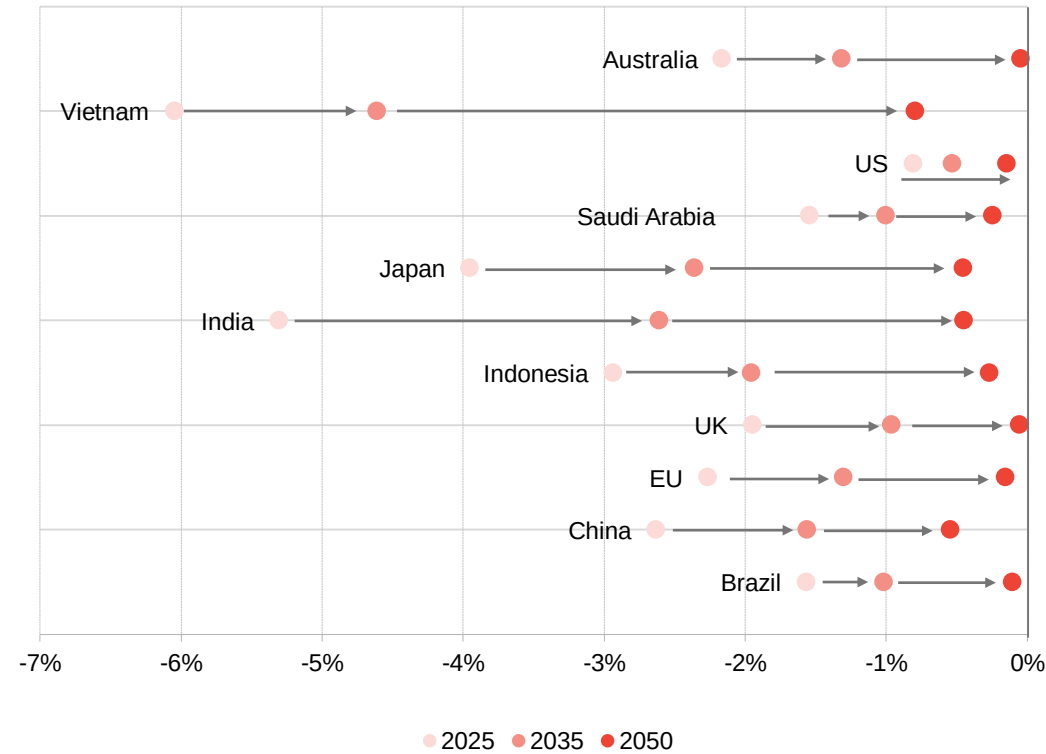
Energy imports as share of GDP (\$, real 2025)



Source: BloombergNEF Trade Transition Scenario Tool, Sinoimex Global Trade Flow (GTF), Global Change Analysis Model (GCAM).

Energy commodity import dependence by market and scenario, 2025, 2035, 2050, NZS

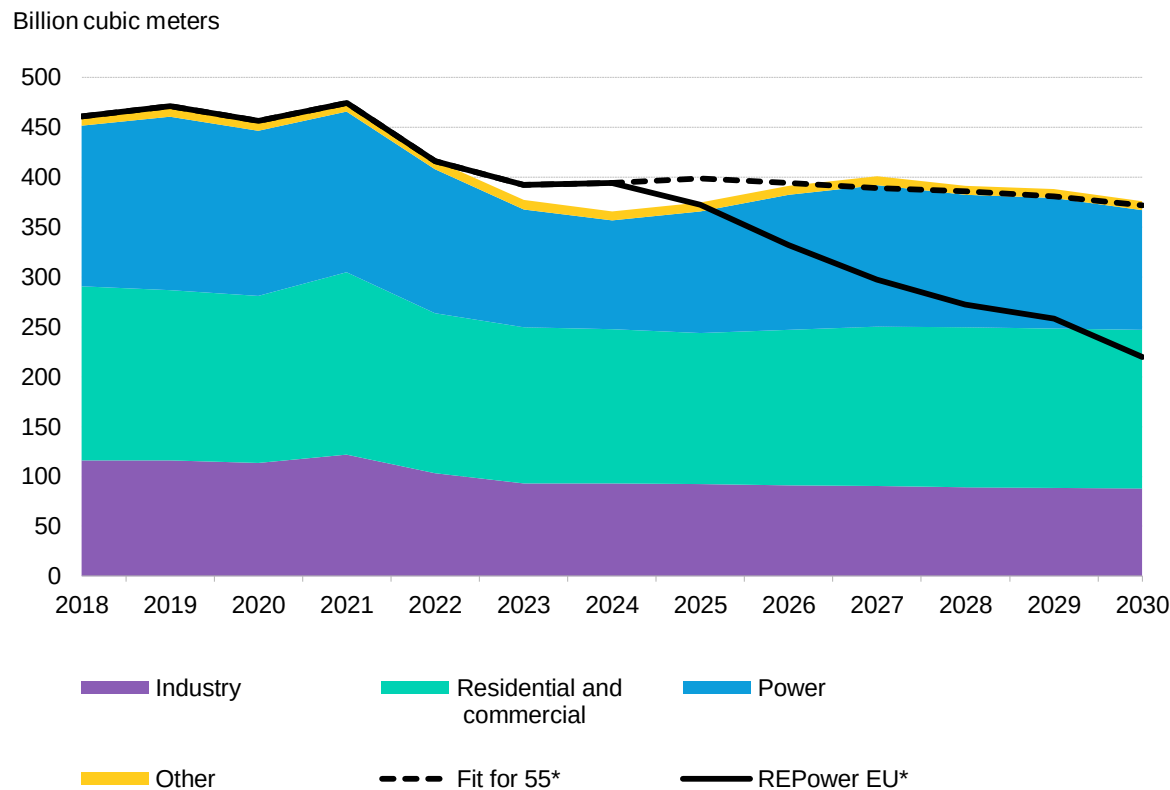
Energy imports as share of GDP (\$, real 2025)



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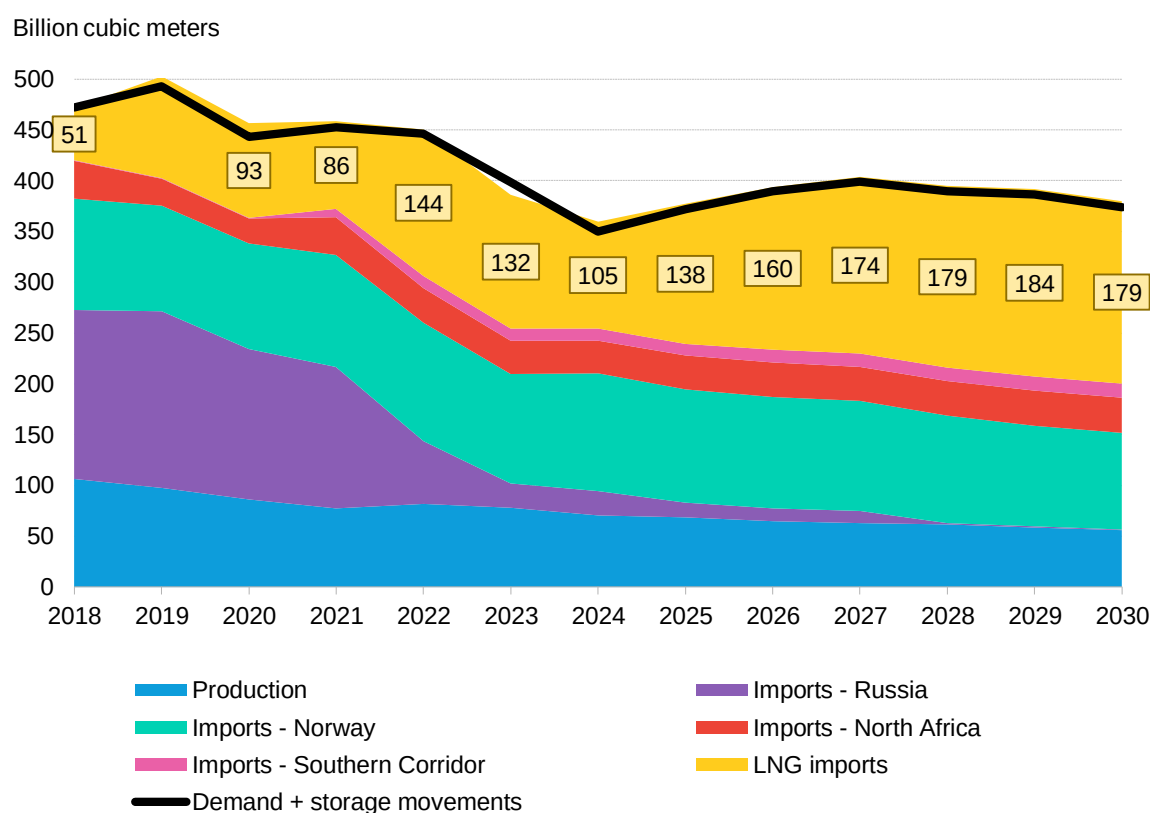
Geopolitical conflict first reshapes fuel flows and risk premiums

BNEF's European gas demand outlook



Source: BloombergNEF European 2030 Gas Market Outlook, as of July 2025, European Network of Transmission System Operators for Gas, transmission system operators.

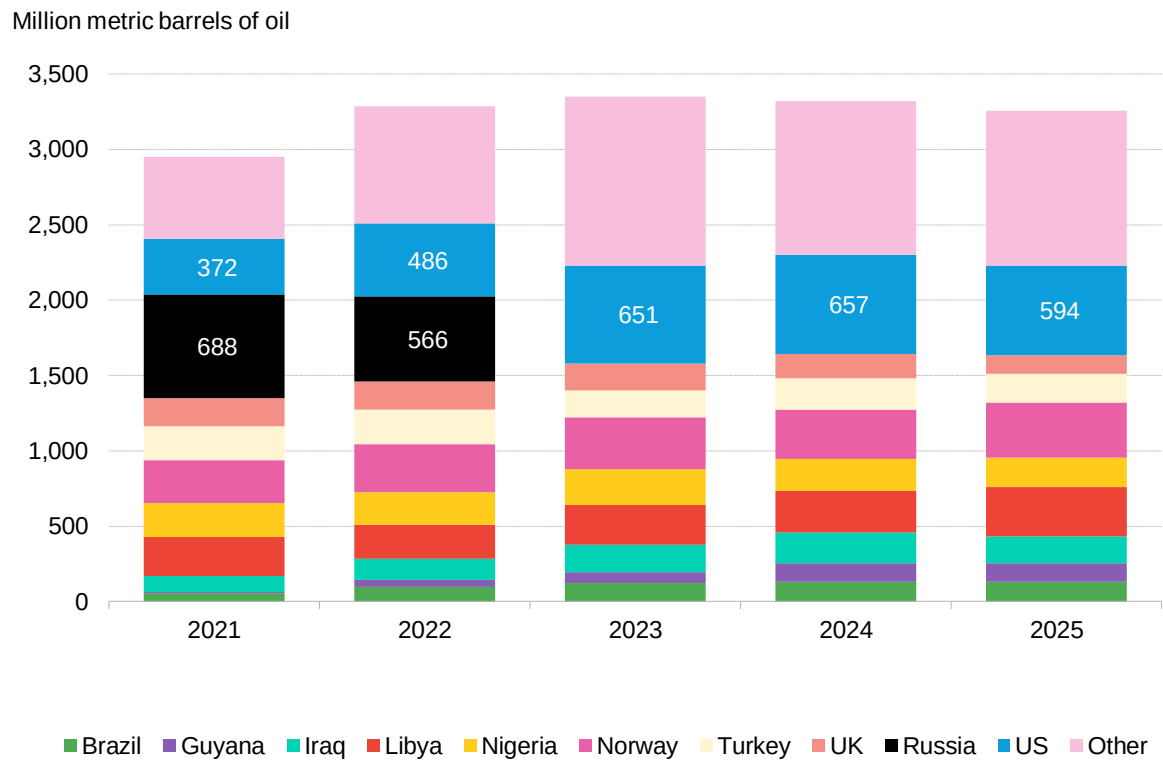
BNEF's European gas supply outlook



Source: BloombergNEF, ENTSOG, Transmission system operators.

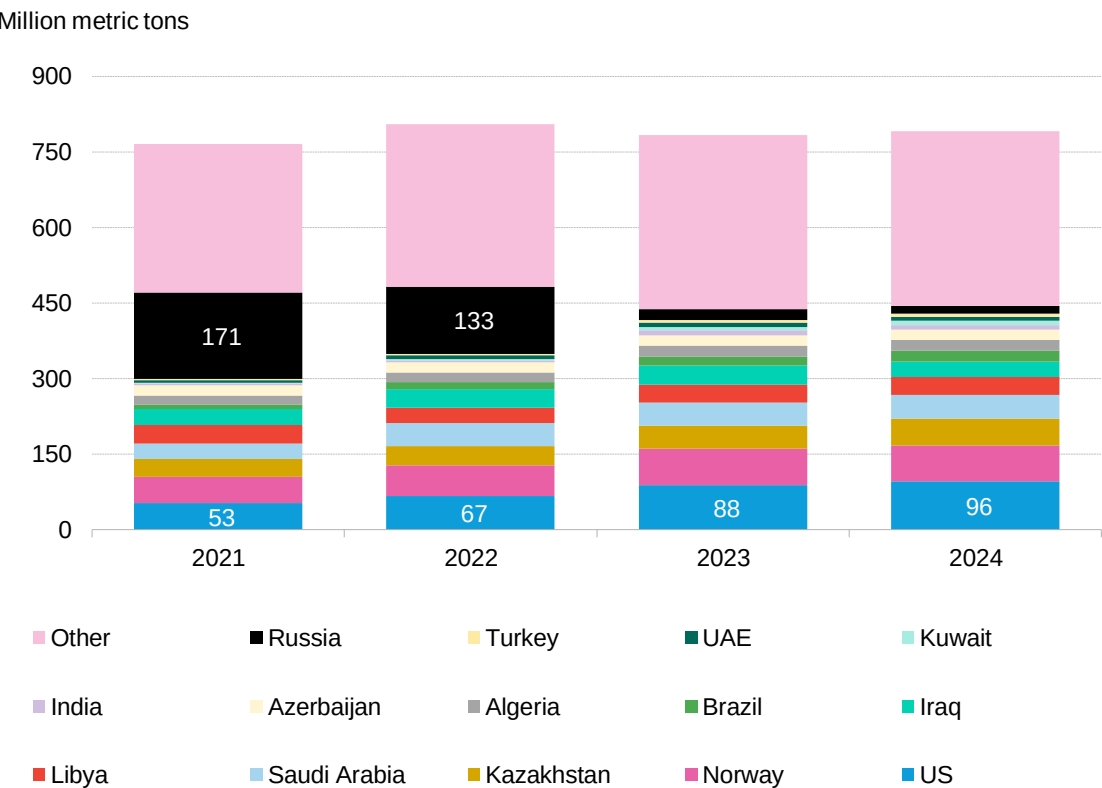
Fuel shocks spread through power costs to the wider economy

Seaborne crude import volumes to Western Europe, by economy of origin.



Source: BloombergNEF, Eurostat

Oil and petroleum product imports to the EU by economy of origin

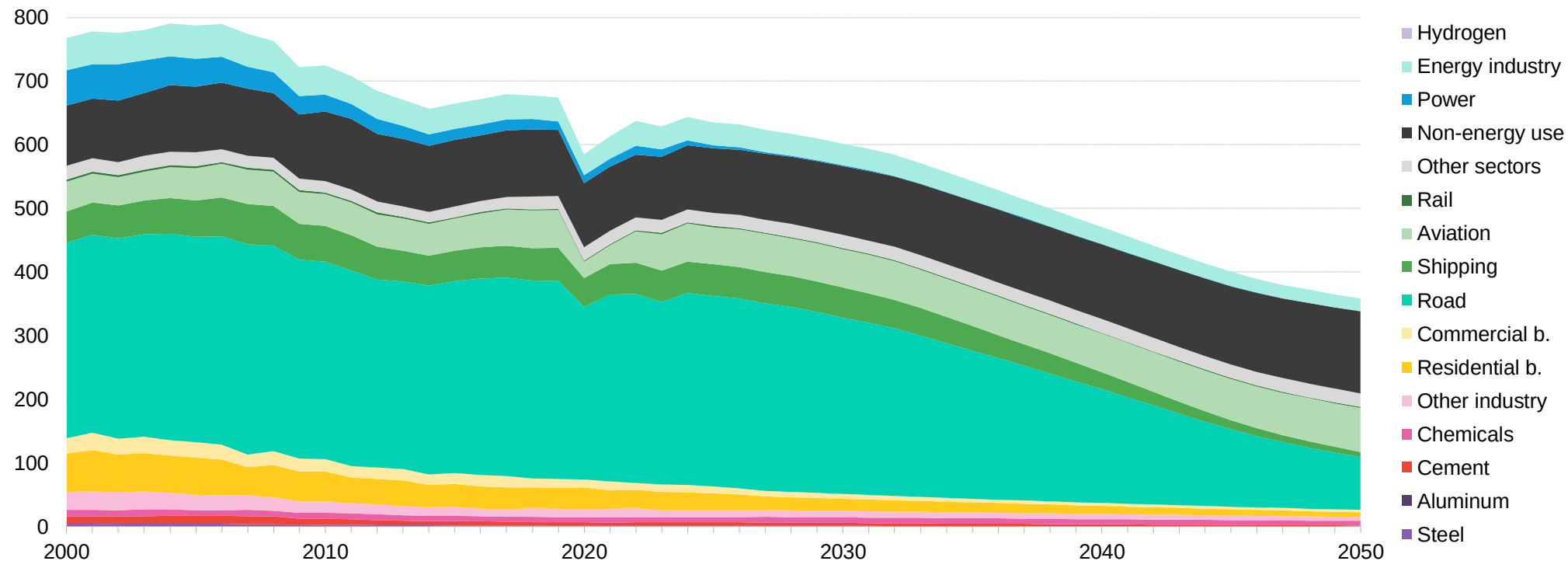


Source: BloombergNEF, Eurostat

Fuel shocks spread through power costs to the wider economy

European oil consumption by sector, Economic Transition Scenario in BNEF's NEO 2025

Million tons of oil equivalent

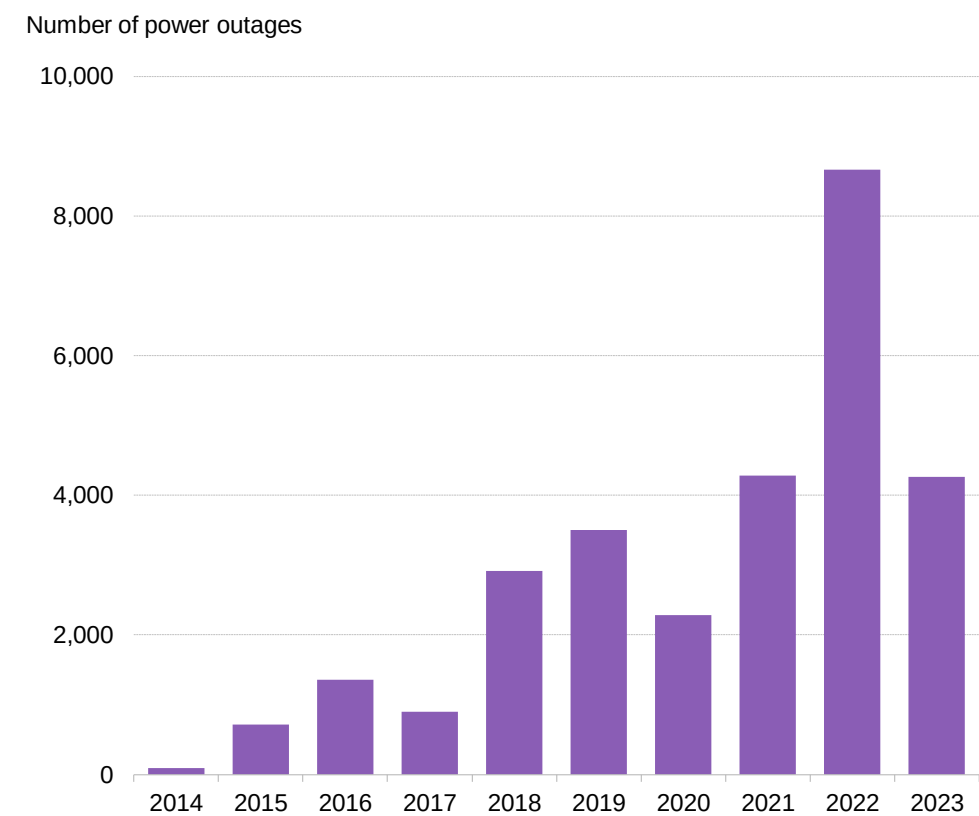


Source: BloombergNEF. Note: Scenario refers to Economic Transition Scenario (ETS) from NEO 2025. Commercial and residential segments refer to buildings.



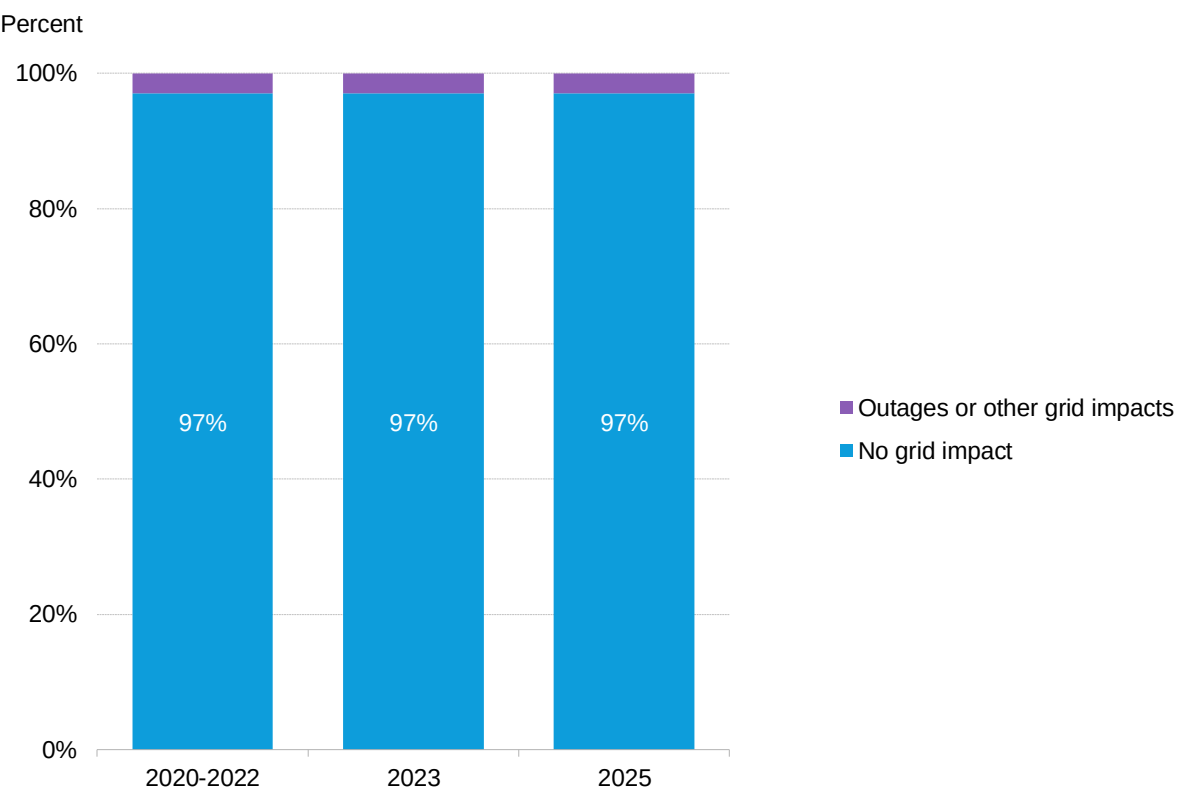
Energy security now extends to the physical resilience of the grid

US county-level power outages by physical attacks



Source: BloombergNEF, Pacific Northwest National Laboratory, Environment for the Analysis of Geo-Located Energy Information, DOE OE-417.

Share of North American grid physical incidents with operational disruption

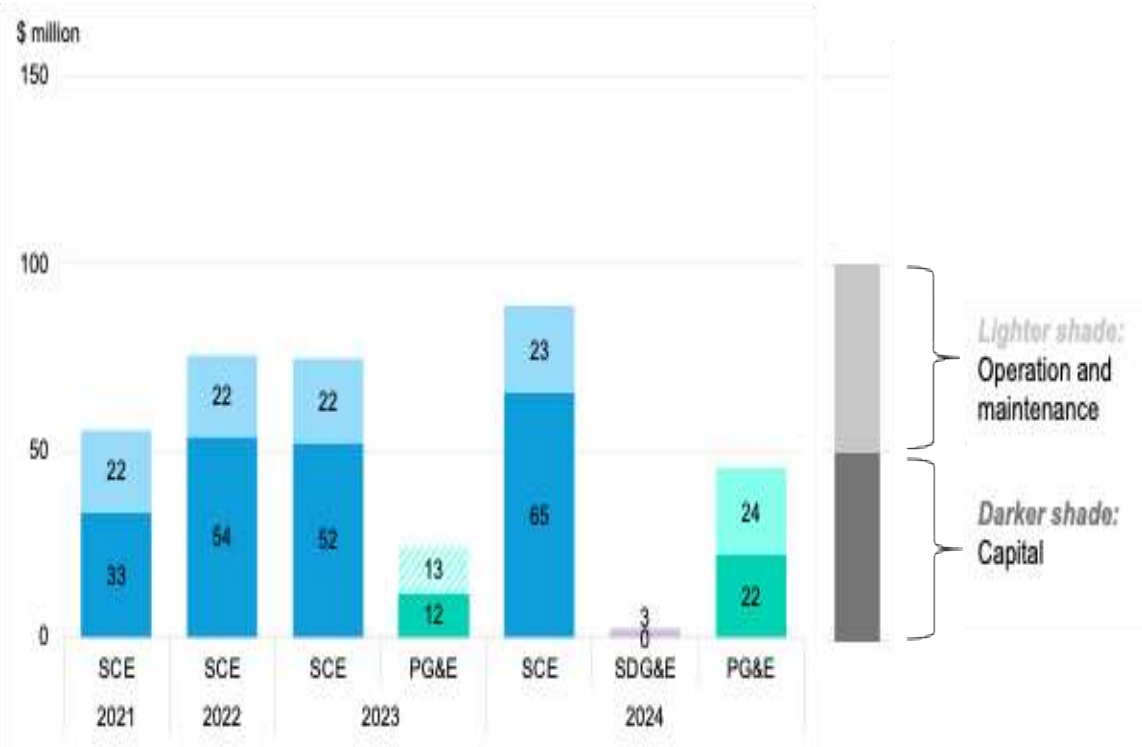


Source: Electricity Information Sharing and Analysis Center, BloombergNEF.



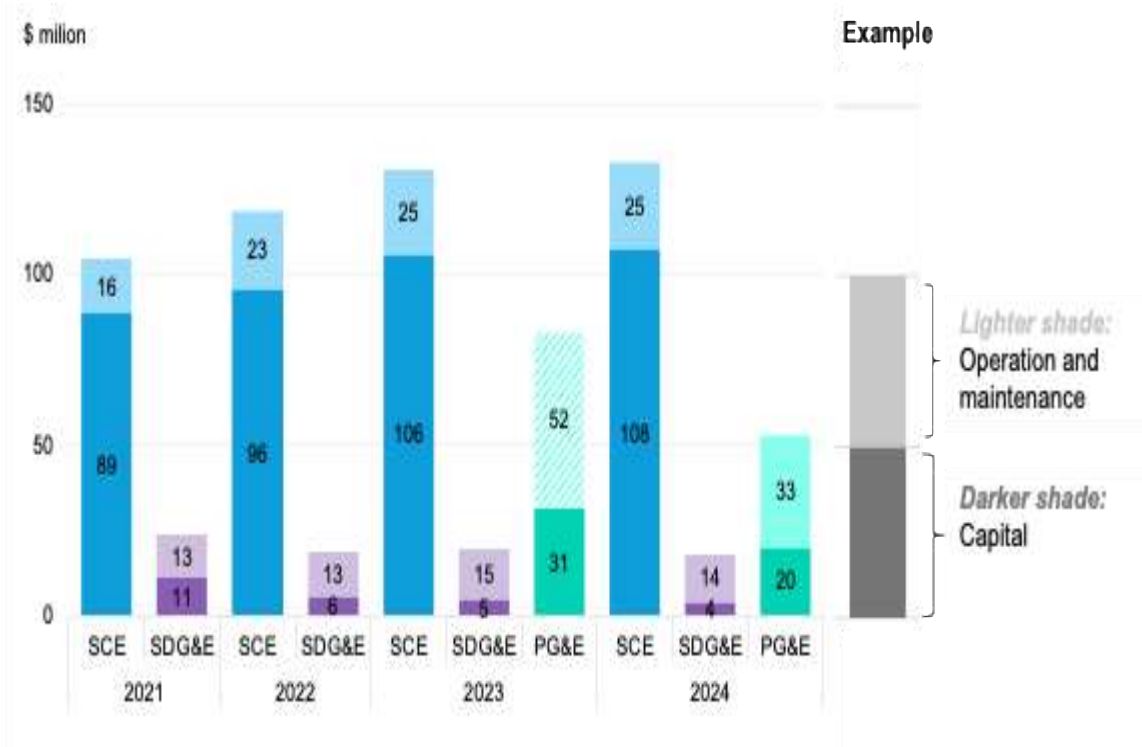
Grid digitalization expands the cyberattack surface and redirects investment

Physical security spend by investor-owned utilities in California



Source: BloombergNEF, CPUC Risk Spending Accountability Reports (RSAR).
Note: Dashed bars indicated estimated values. SCE is Southern California Edison, SDG&E is San Diego Gas & Electric, PG&E is Pacific Gas & Electric.

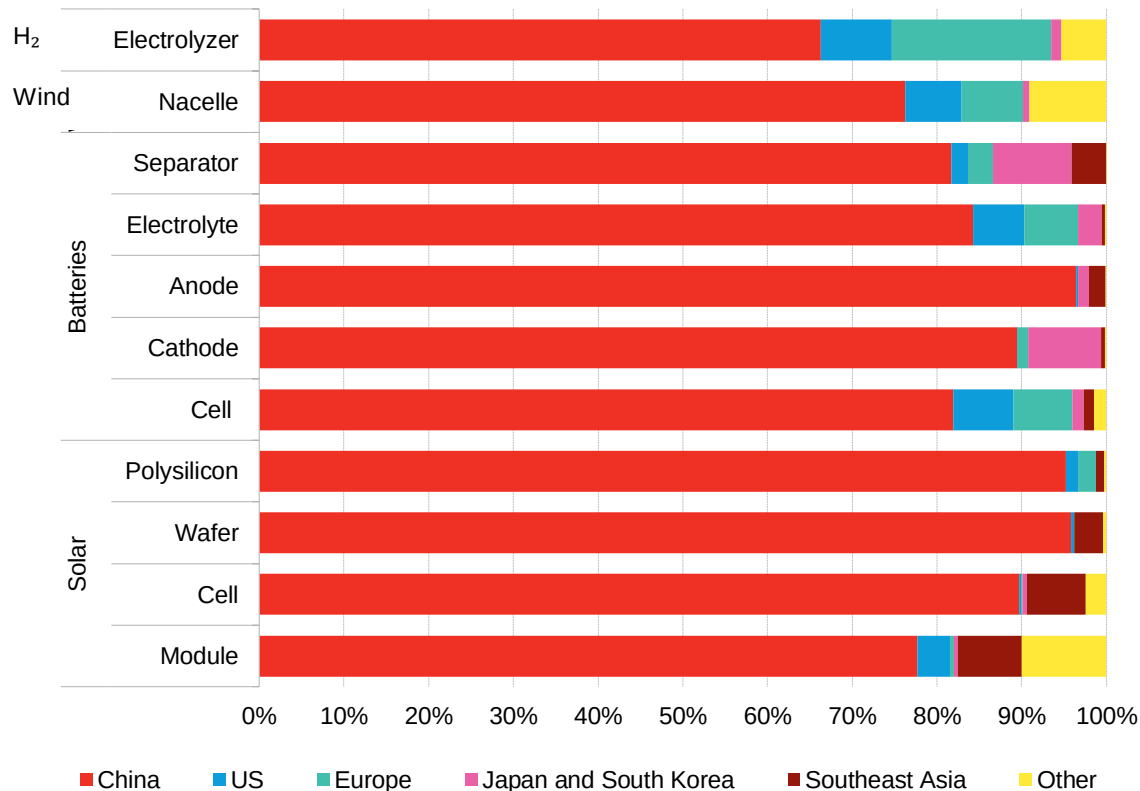
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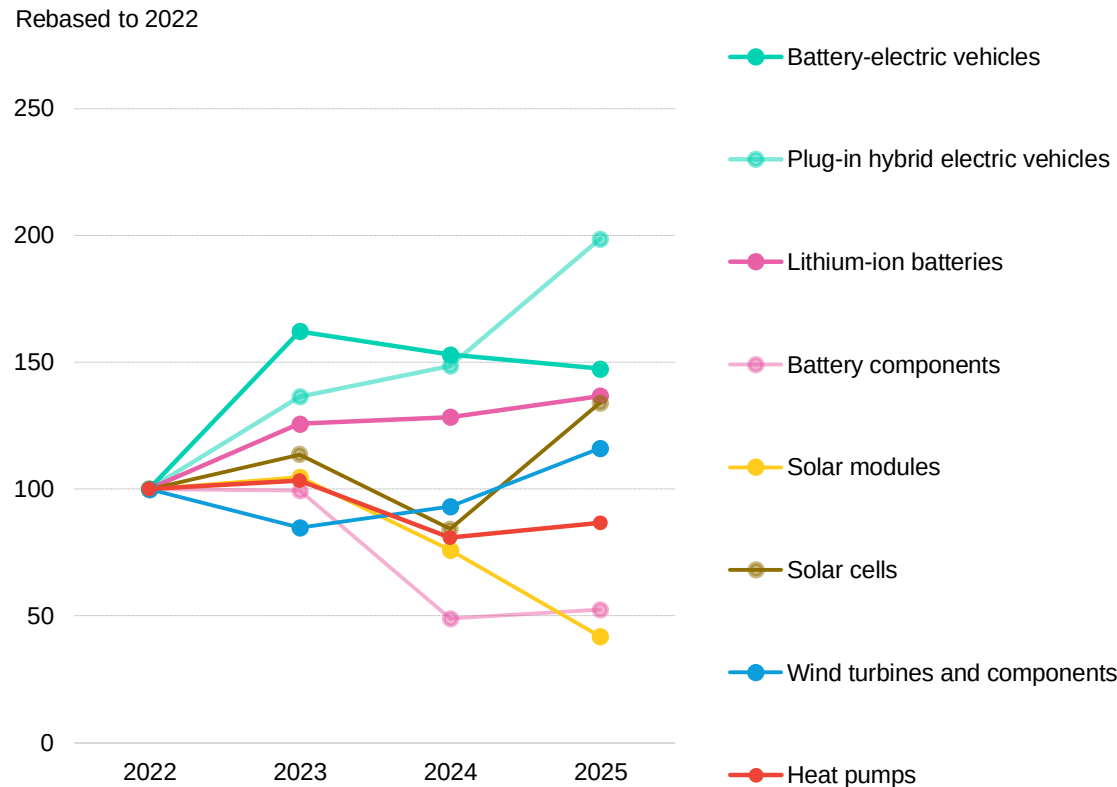
Clean-tech supply chains are shifting from lowest cost to security and diversification

Clean energy manufacturing capacity by location in 2025



Source: BloombergNEF

Global imports of clean-tech goods by value



Source: BloombergNEF, Sinoimex.

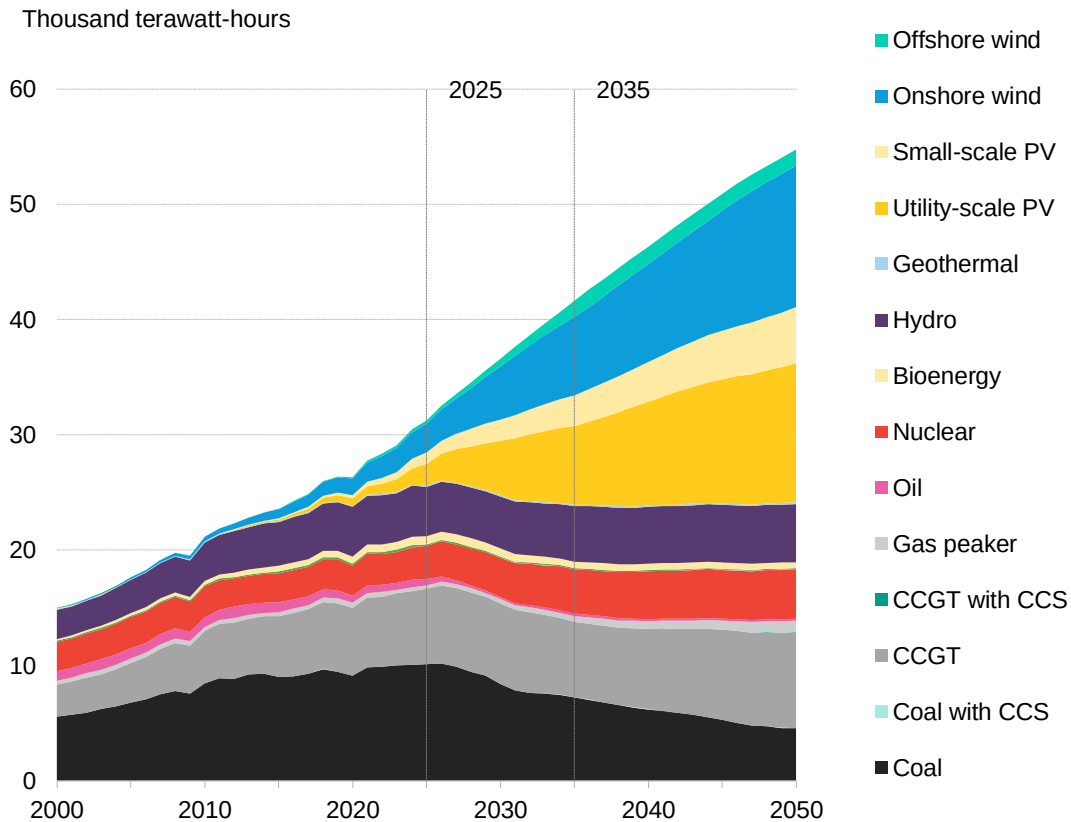
TWO

THE ENERGY TRANSITION IS RESHAPING POWER SUPPLY



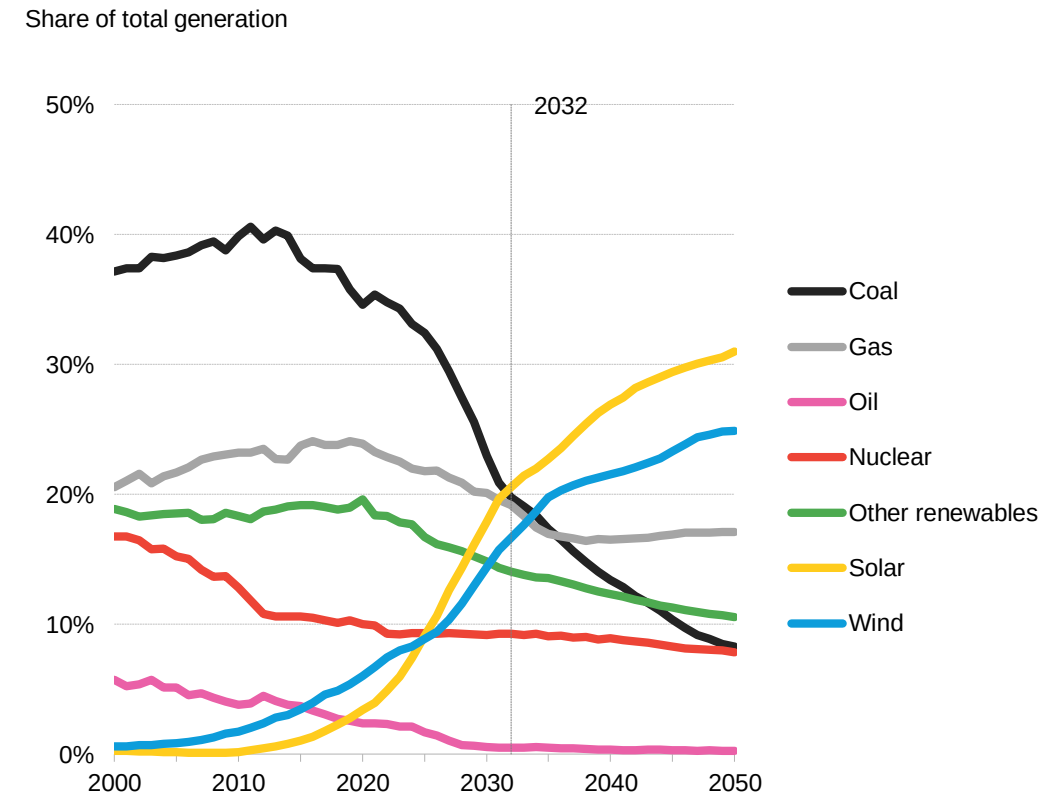
Low-marginal-cost renewables are rewriting the merit order

Electricity generation by technology, Economic Transition Scenario



Source: BloombergNEF

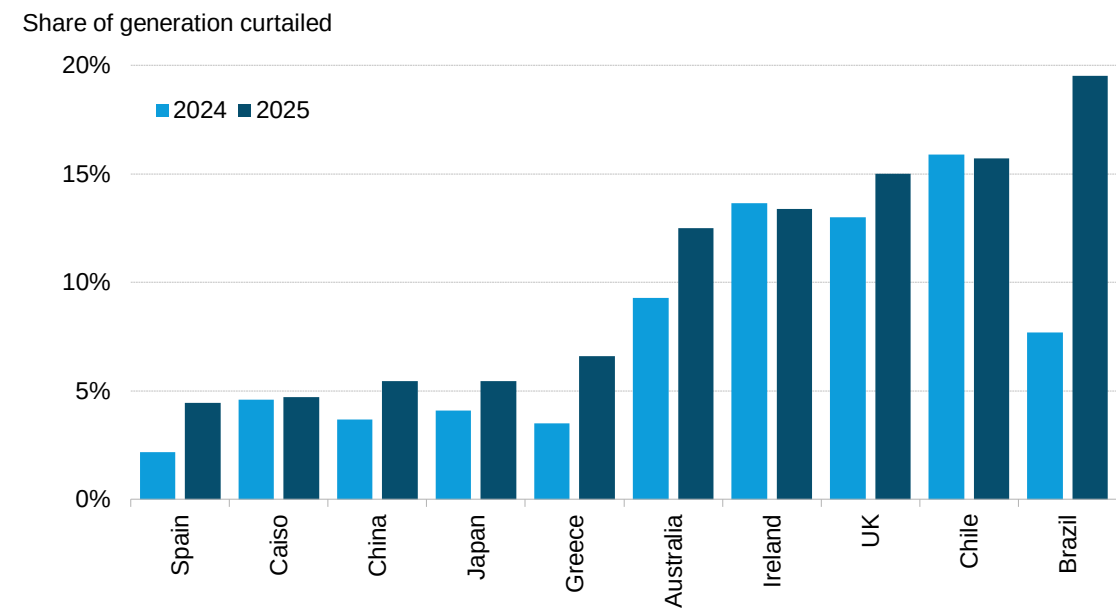
Global electricity generation by technology, Economic Transition Scenario



Source: BloombergNEF

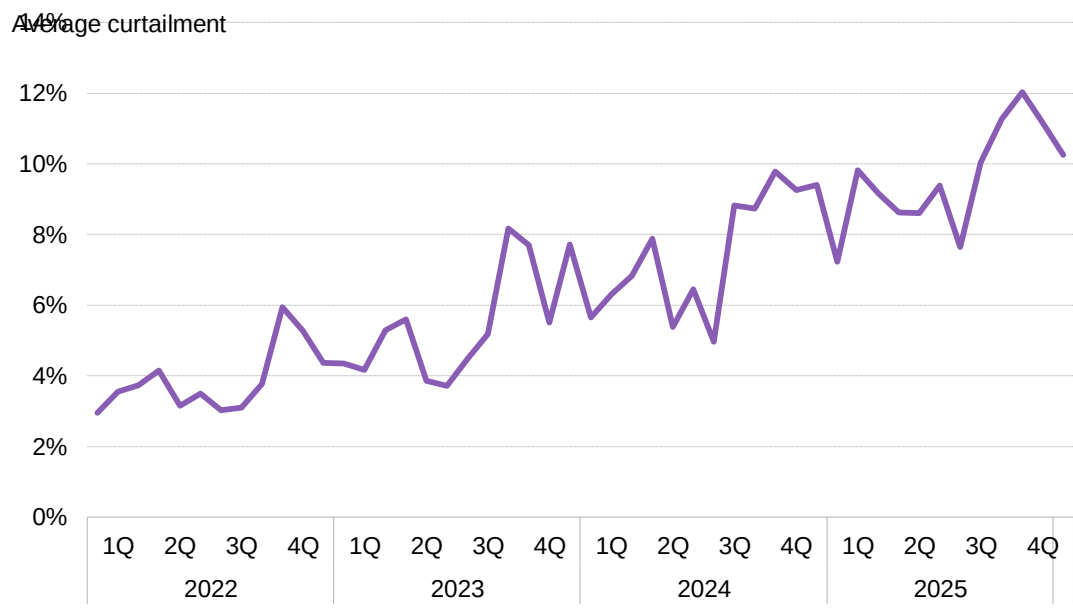
Concentrated renewable output erodes its own capture price

Total wind and solar curtailment rates in 2024 and 2025, by market



Source: BloombergNEF, Red Electrica, California Independent System Operator (Caiso), China, National Energy Administration, Japan Regional Electric Utilities, Independent Power Transmission Operator (ADMIE), Australian Energy Market Operator, EirGrid, National Energy System Operator, Coordinador Eléctrico Nacional, Operador Nacional do Sistema Eléctrico.

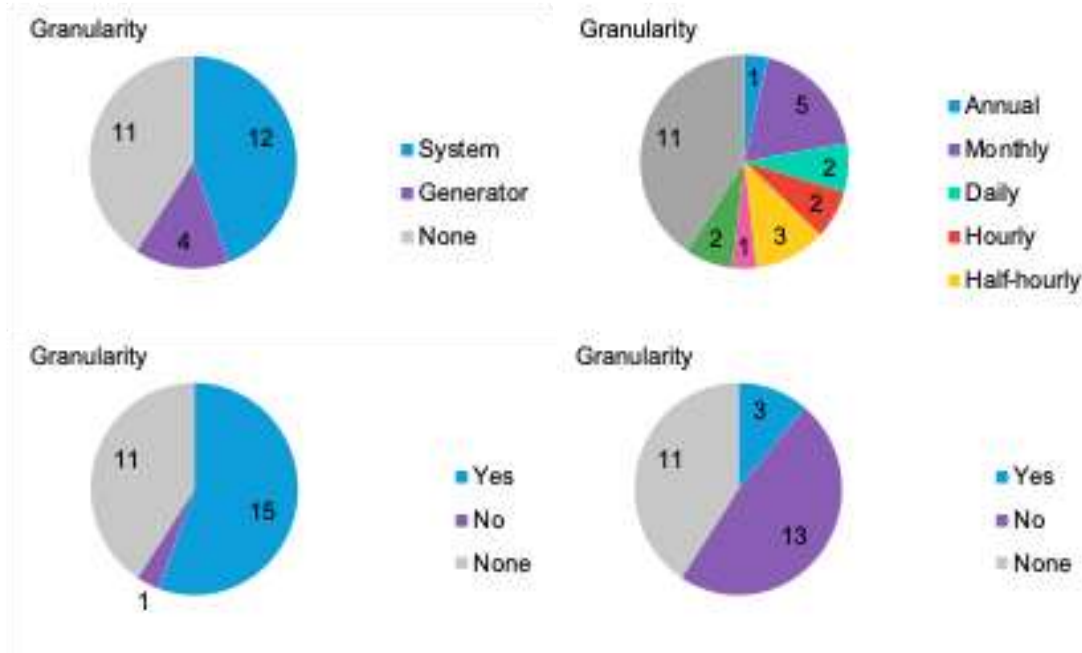
Average curtailment of wind and solar in nine key markets with data since 2022



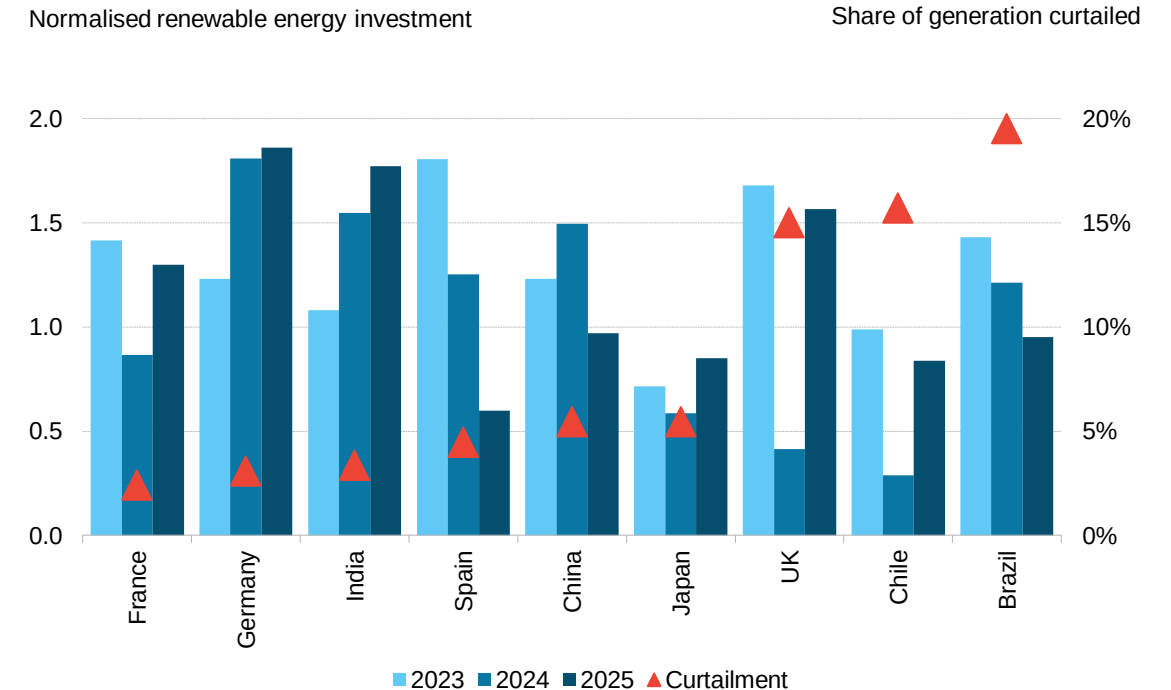
Source: BloombergNEF, Red Electrica, California Independent System Operator (Caiso), China, National Energy Administration, Japan Regional Electric Utilities, Independent Power Transmission Operator (ADMIE), Australian Energy Market Operator, EirGrid, National Energy System Operator, Coordinador Eléctrico Nacional, Operador Nacional do Sistema Eléctrico.

Renewable curtailment reflects multiple system constraints, not a single cause

How curtailment is reported in the 27 markets analyzed by BNEF



Curtailment and renewable energy investment

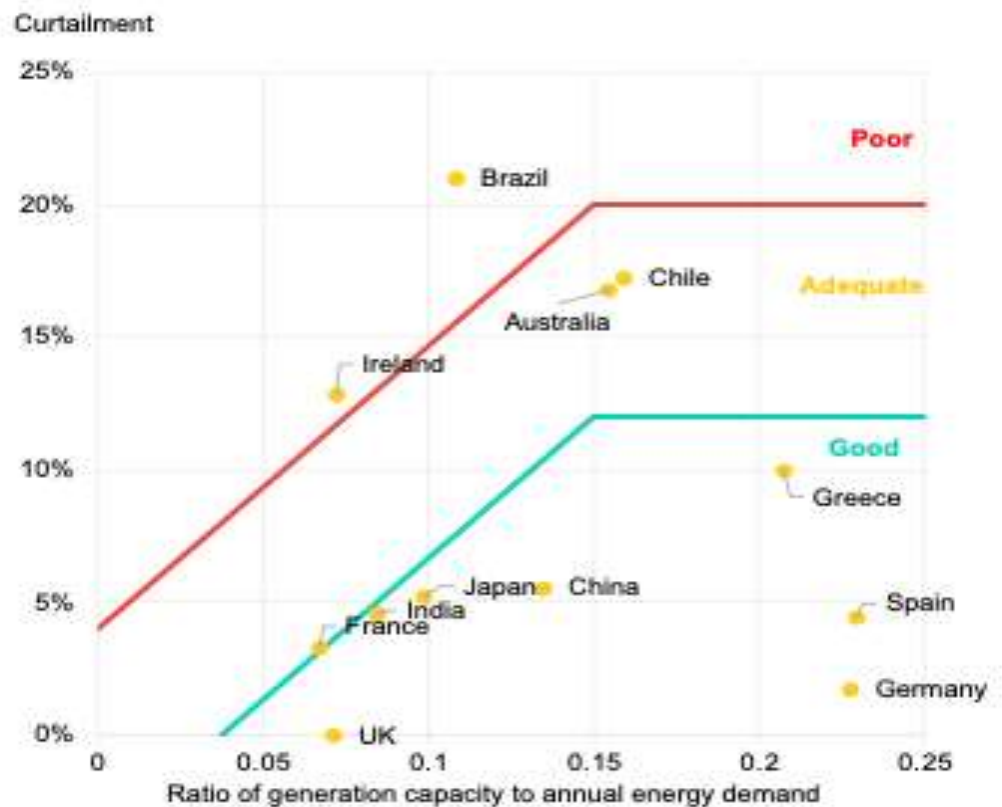


Source: BloombergNEF, California Independent System Operator (CAISO), Australian Energy Market Operator, Red Electrica, National Energy System Operator, China National Energy Administration, Operador Nacional do Sistema Eléctrico, Coordinador Eléctrico Nacional, Kyushu Electric, Shikoku Electric, Chugoku Electric, Kansai Electric, Hokuriku Electric, Chubu Electric, Tokyo Electric, Tohoku Electric, Hokkaido Electric, Bundesnetzagentur, Cyprus Energy Regulatory Authority, EirGrid, Independent Power Transmission Operator (ADMIE), Grid Controller of India

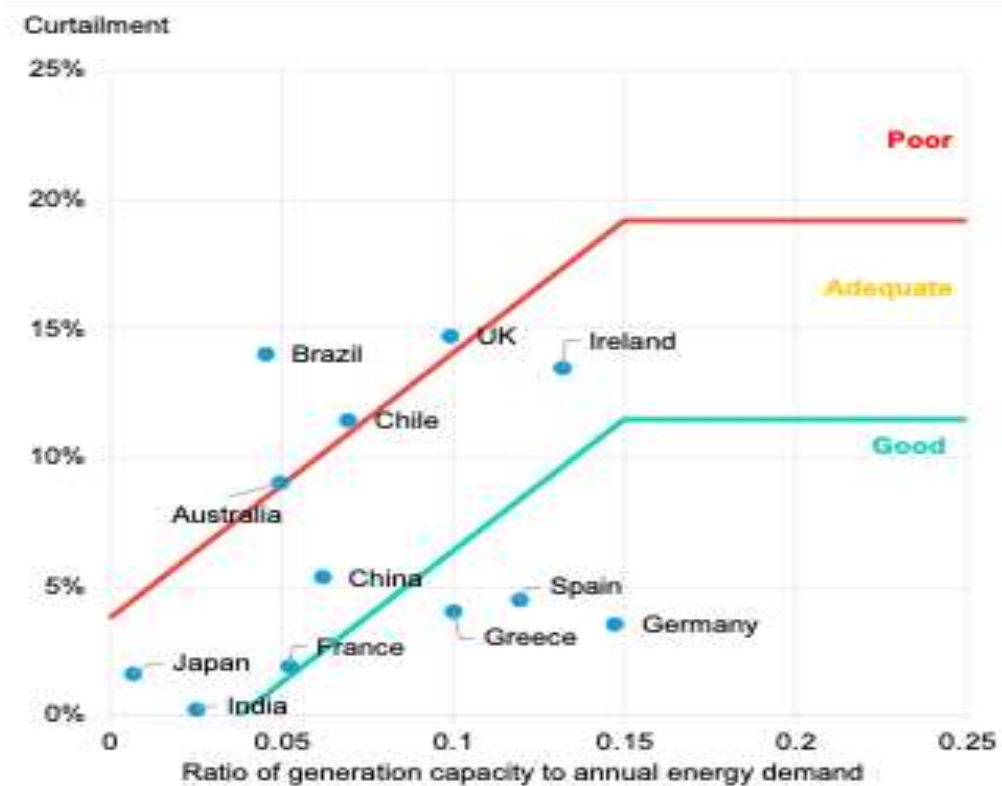
At high renewable penetration, the bottleneck shifts from deployment to integration

Solar generation curtailment relative to solar penetration for selected markets

Wind generation curtailment relative to wind penetration for selected markets



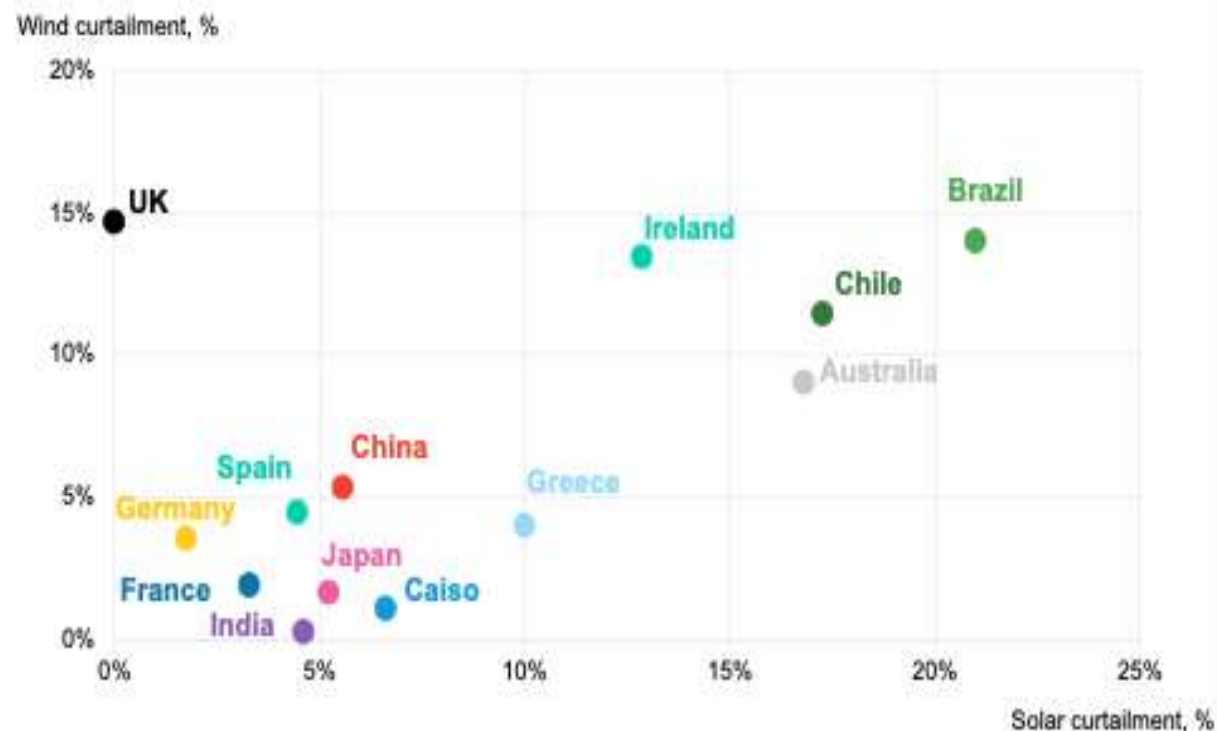
Source: BloombergNEF, power system and market operator reporting



Source: BloombergNEF, power system and market operator reporting

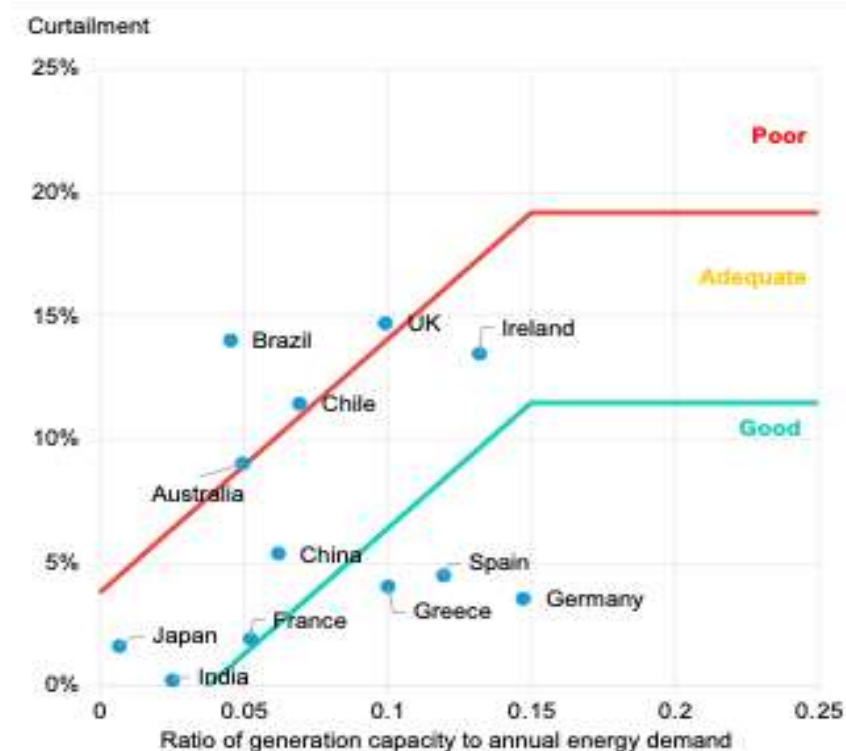
At high renewable penetration, the bottleneck shifts from deployment to integration

2025 wind curtailment versus solar curtailment, by market



Source: BloombergNEF, power system and market operator reporting

Wind generation curtailment relative to wind penetration for selected markets

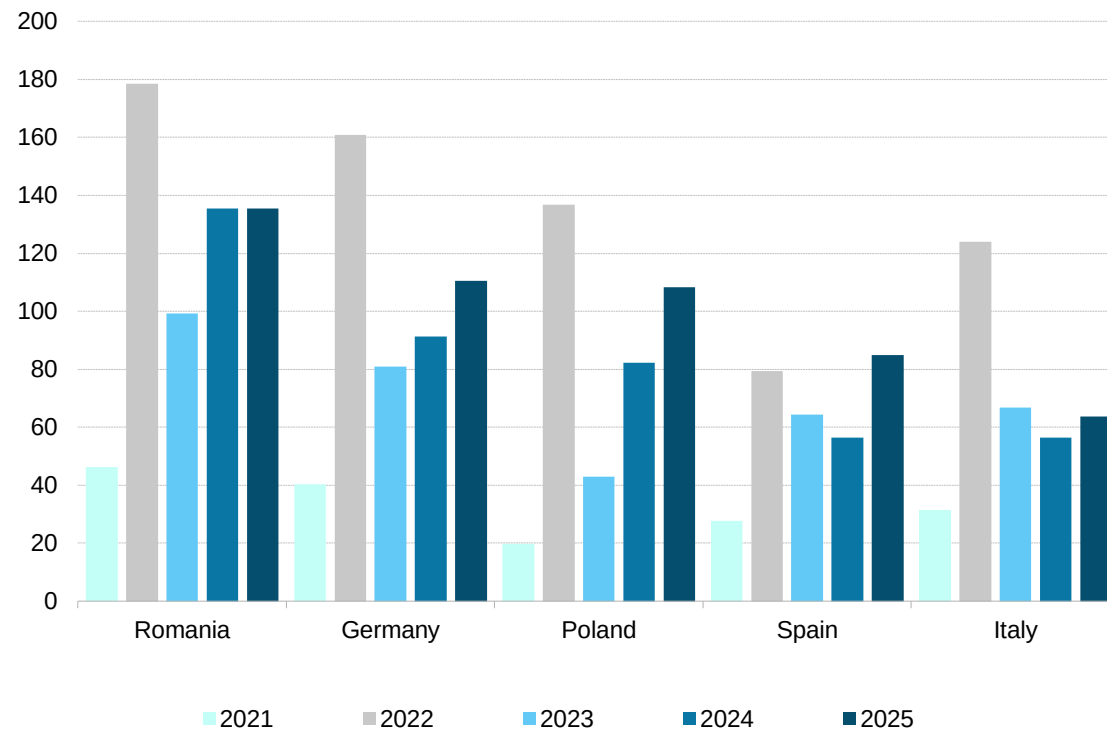


Source: BloombergNEF, power system and market operator reporting

Time-specific value matters more than average power prices

Four-hour power price spreads in select European markets, 2022-25

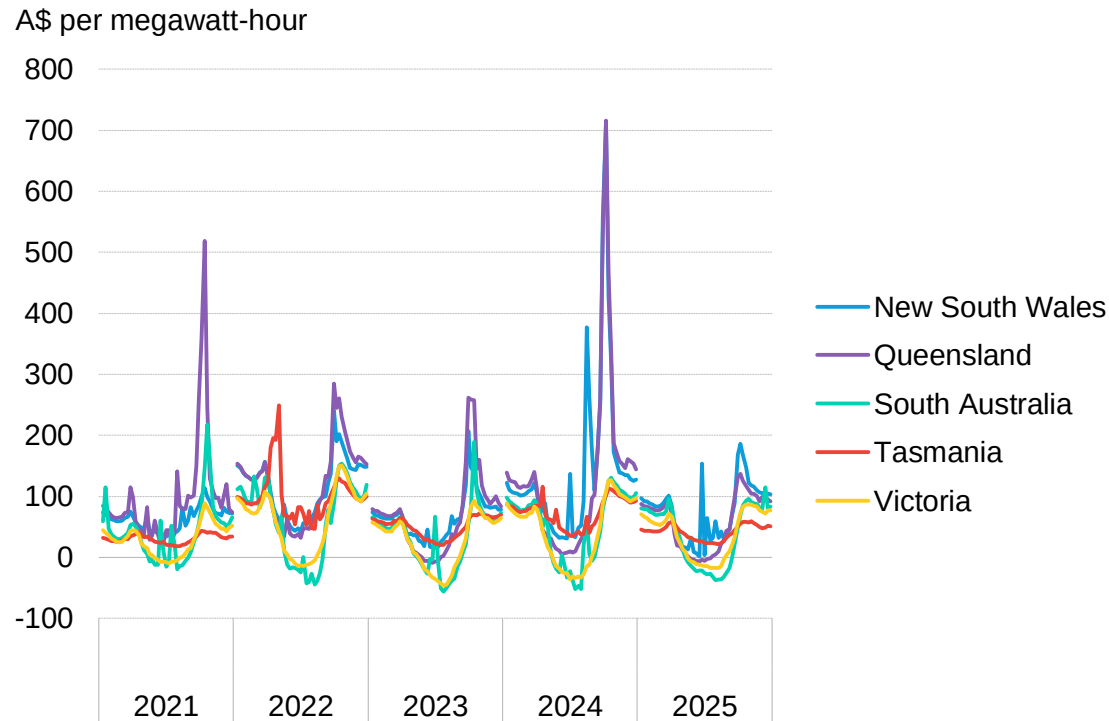
€ per megawatt-hour



Source: BloombergNEF, Bloomberg Terminal (LINE GO)

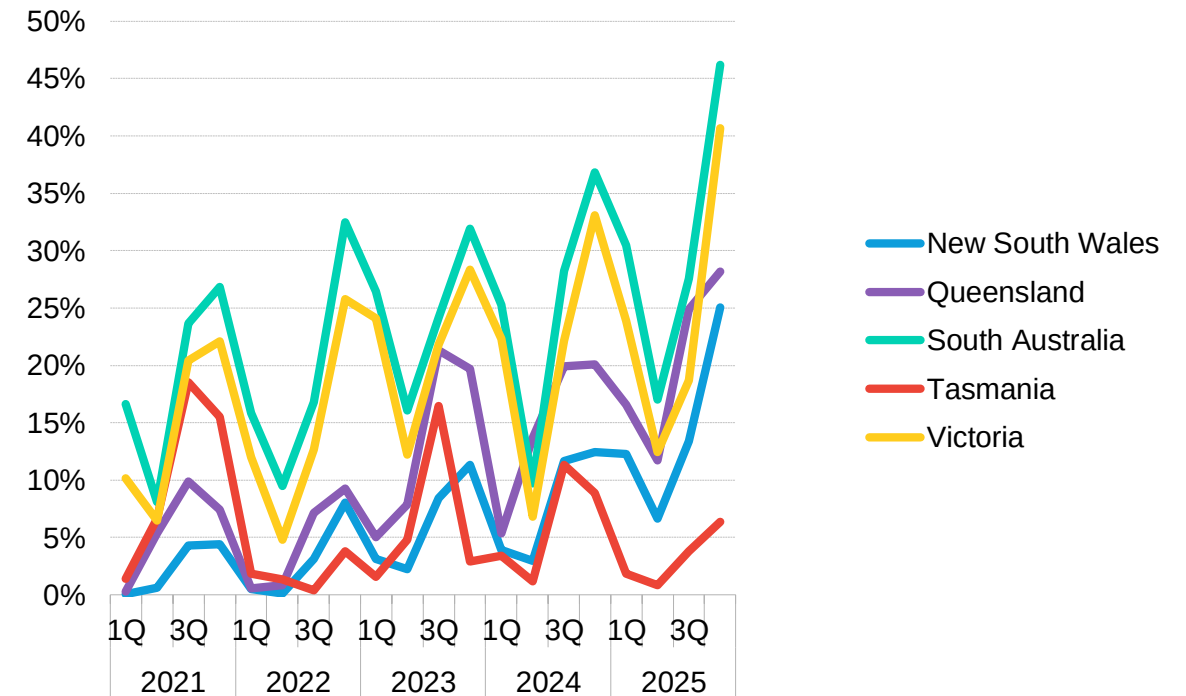
Time-specific value matters more than average power prices

Australia average intraday half-hourly power price profile



Source: Australian Energy Market Operator, BloombergNEF

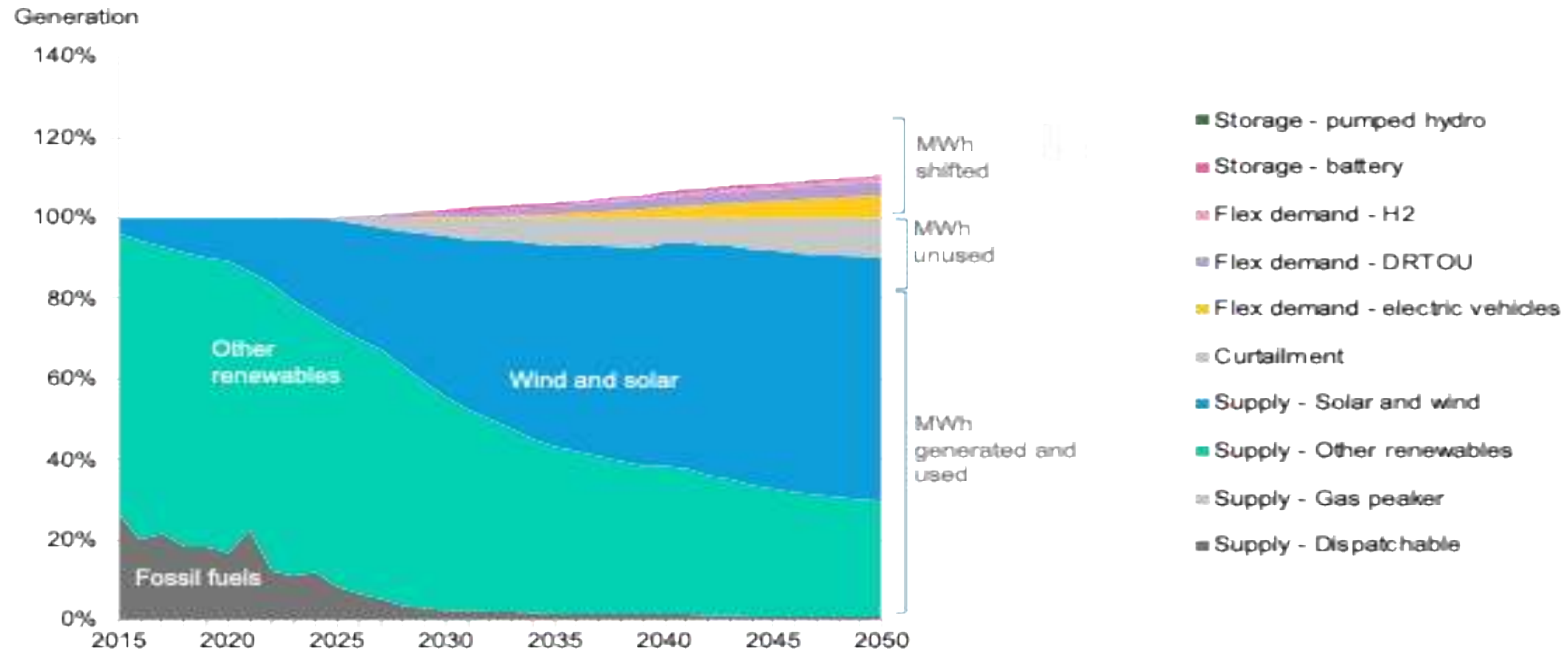
Proportion of negative power prices in the NEM



Source: Australian Energy Market Operator, BloombergNEF

Different flexibility resources address different timescales

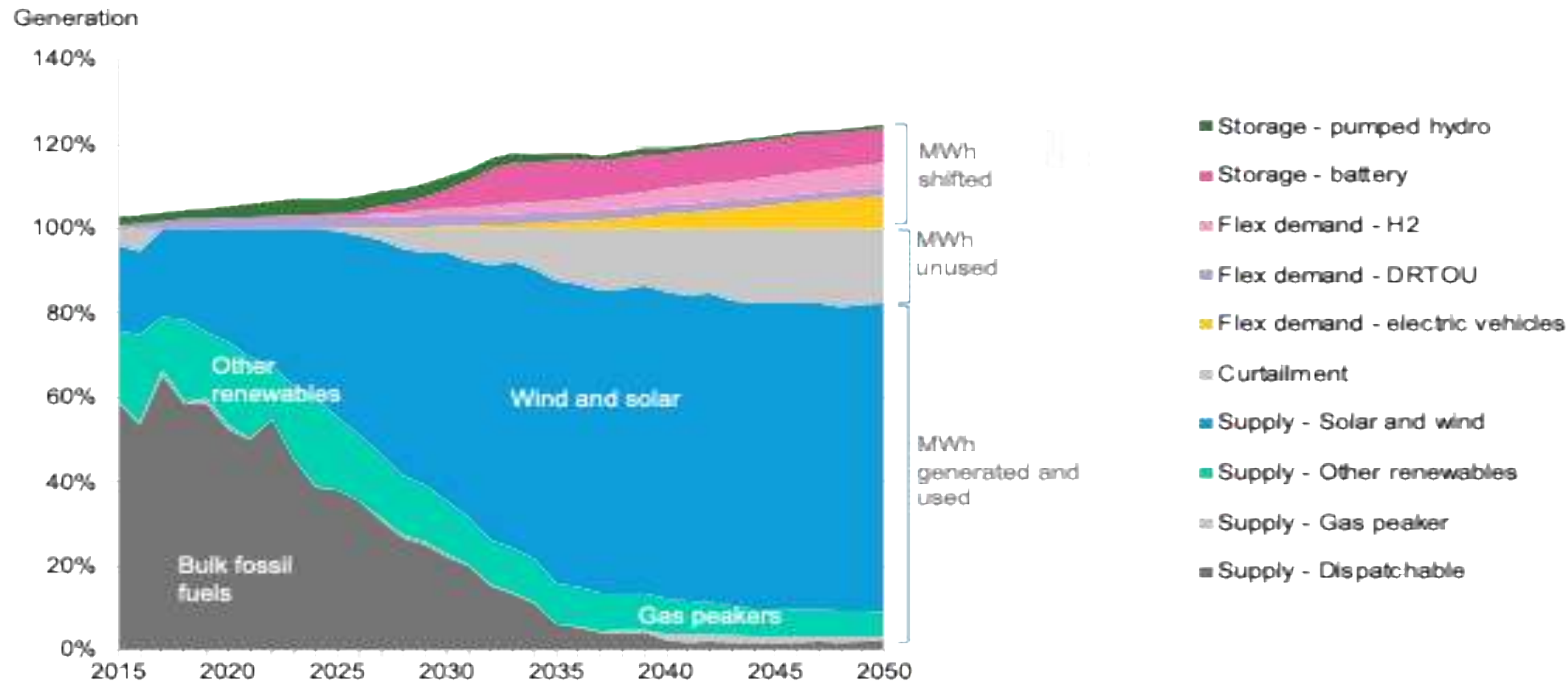
Power generation and power flexibility in Brazil, Economic Transition Scenario



Source: BloombergNEF

Different flexibility resources address different timescales

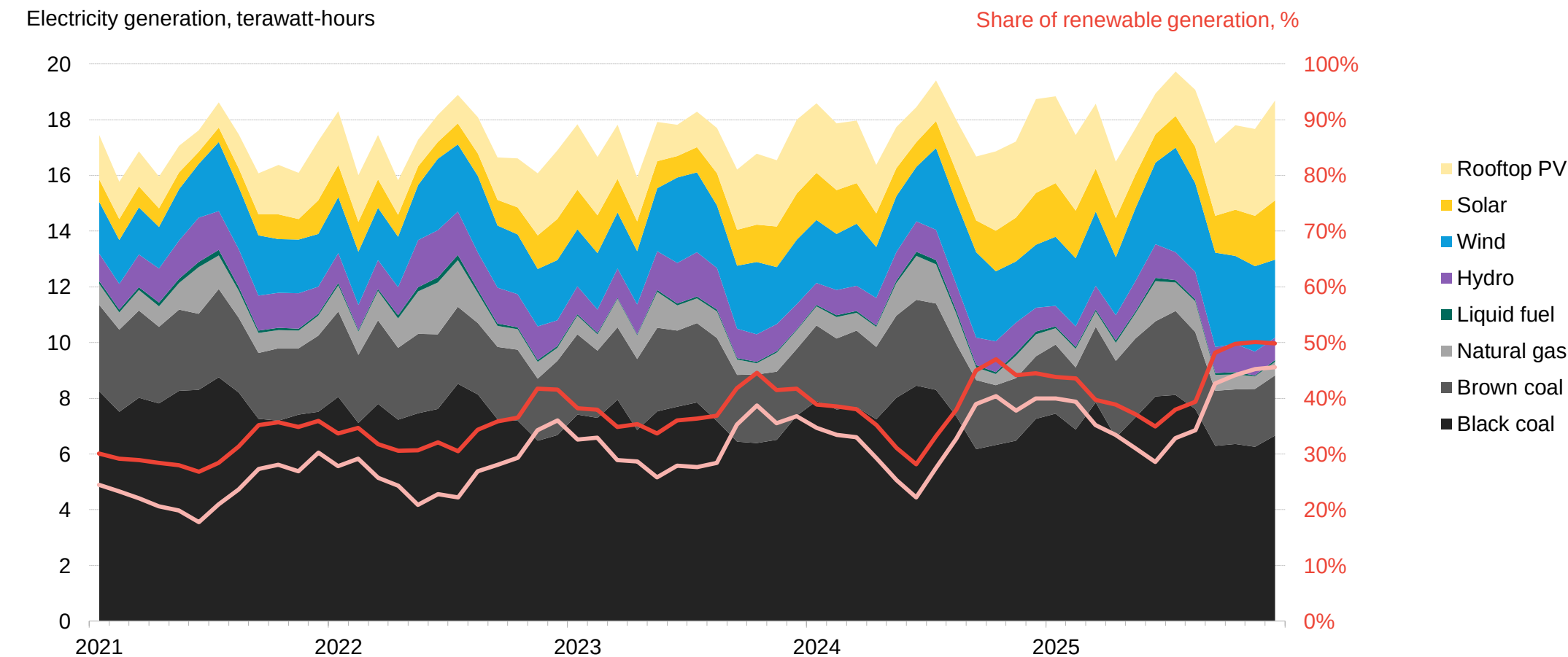
Power generation and power flexibility in Spain and Portugal, Economic Transition Scenario



Source: BloombergNEF

Australia offers an early view of a high-renewables power market

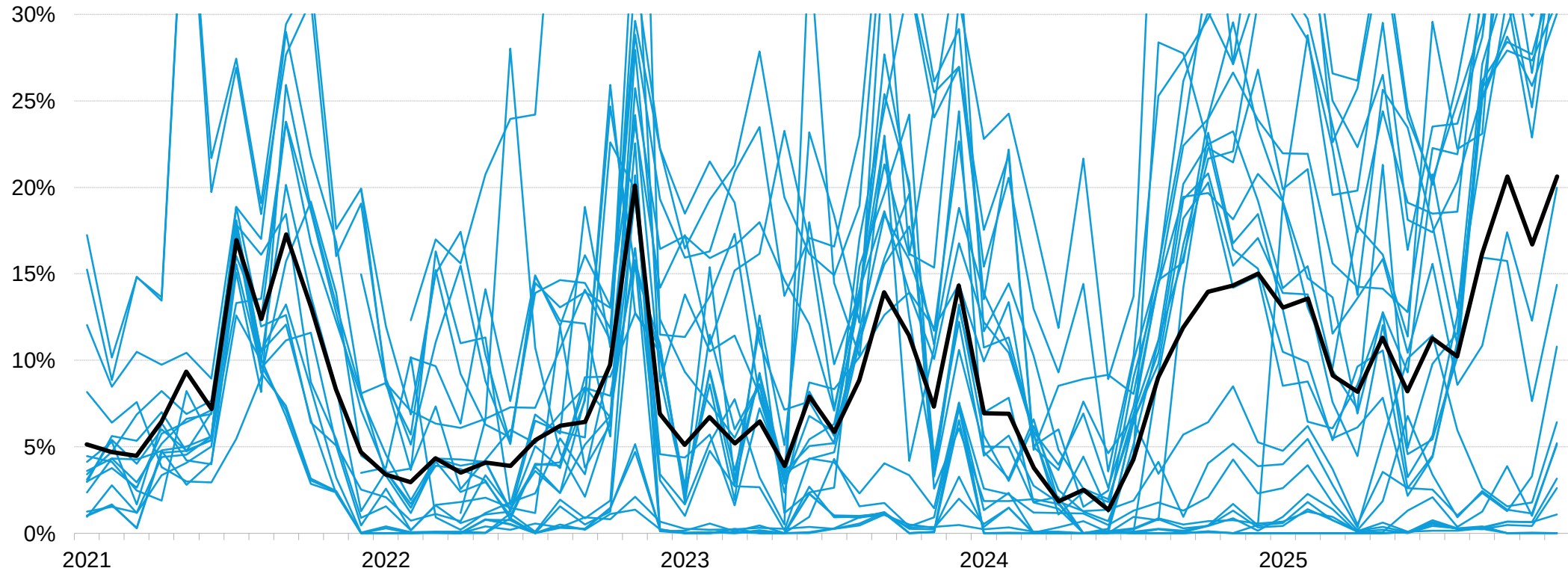
Monthly electricity generation in the NEM, by source



Australia offers an early view of a high-renewables power market

Monthly curtailment for wind generators in South Australia

Curtailment

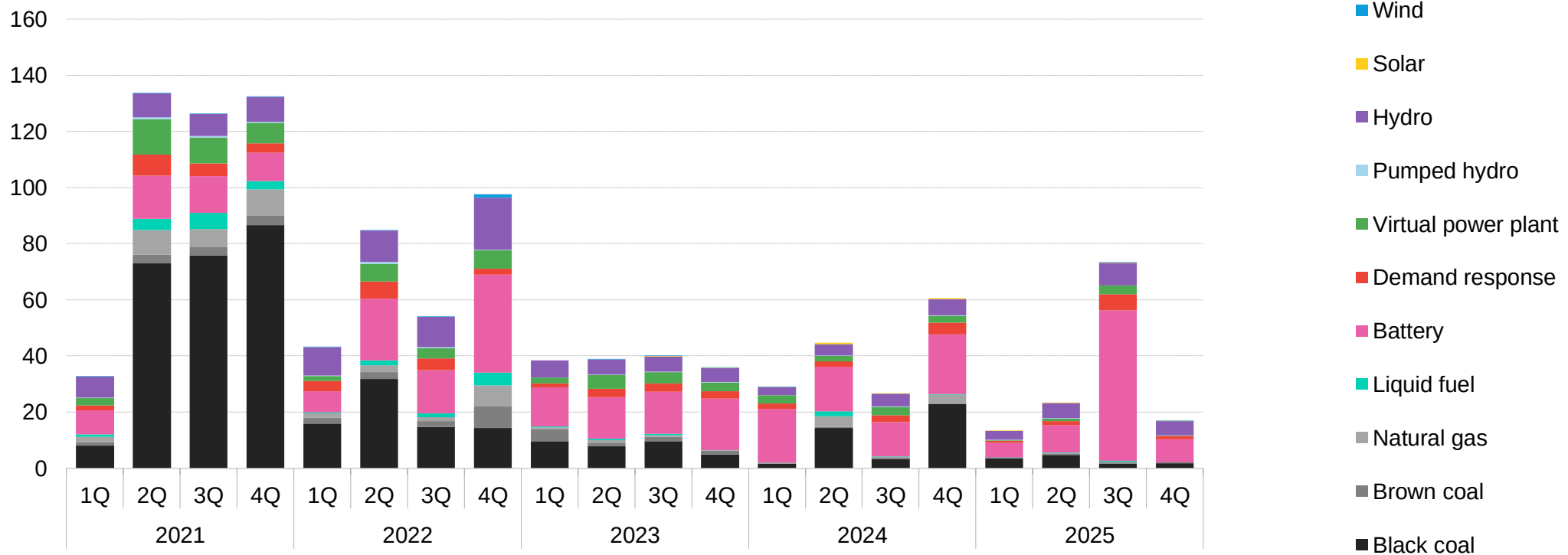


Source: Australian Energy Market Operator, BloombergNEF

Battery revenues reward response at the right time, not more generation

Frequency control ancillary services revenue, by generator type

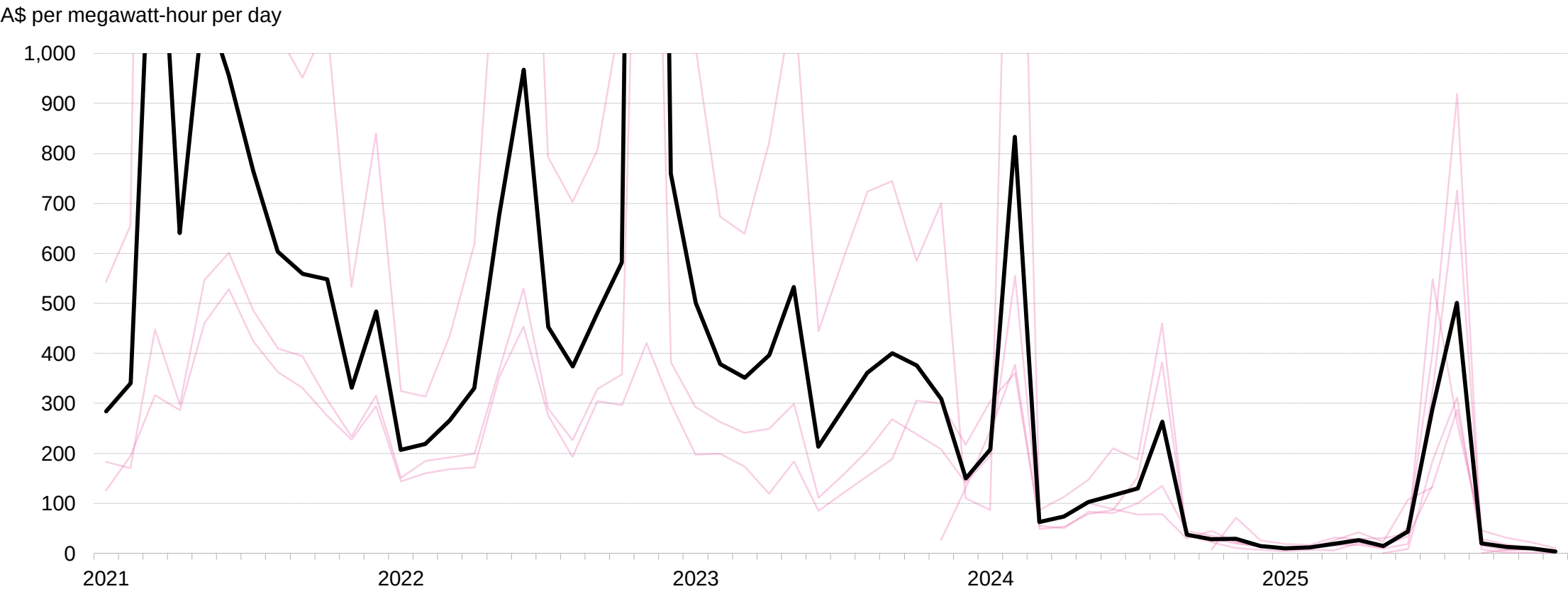
A\$ million



Source: Australian Energy Market Operator, BloombergNEF

Battery revenues reward response at the right time, not more generation

Monthly market value of FCAS for batteries in South Australia



Source: Australian Energy Market Operator, BloombergNEF

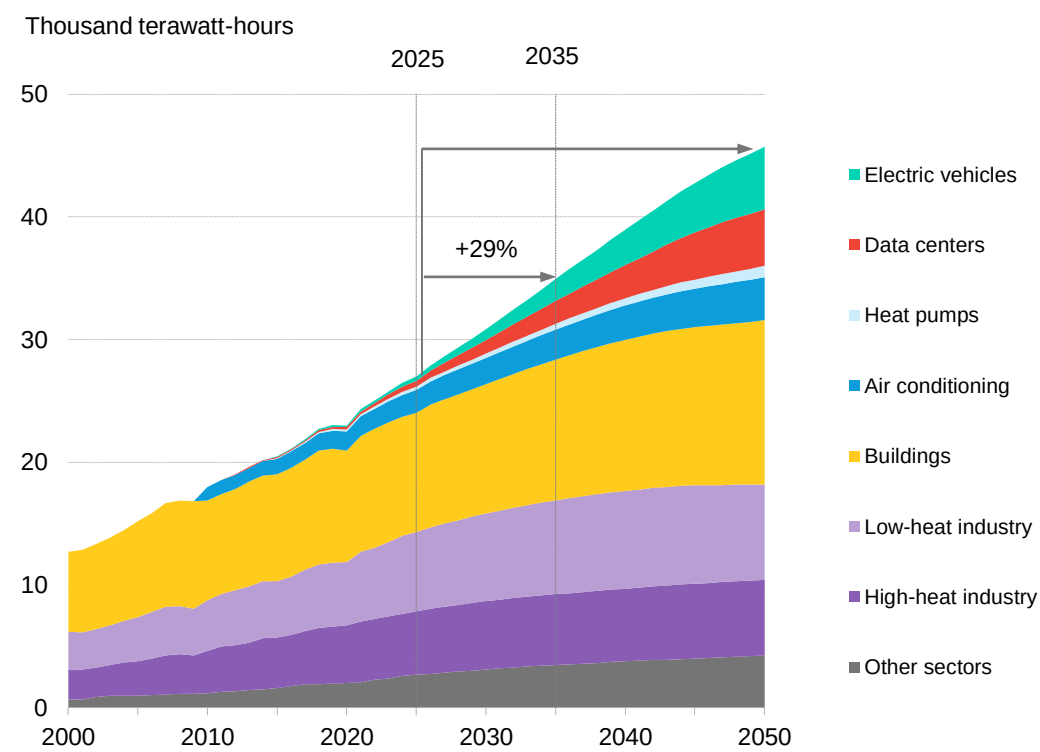
THREE

DATA CENTERS ARE
RESHAPING ELECTRICITY
DEMAND



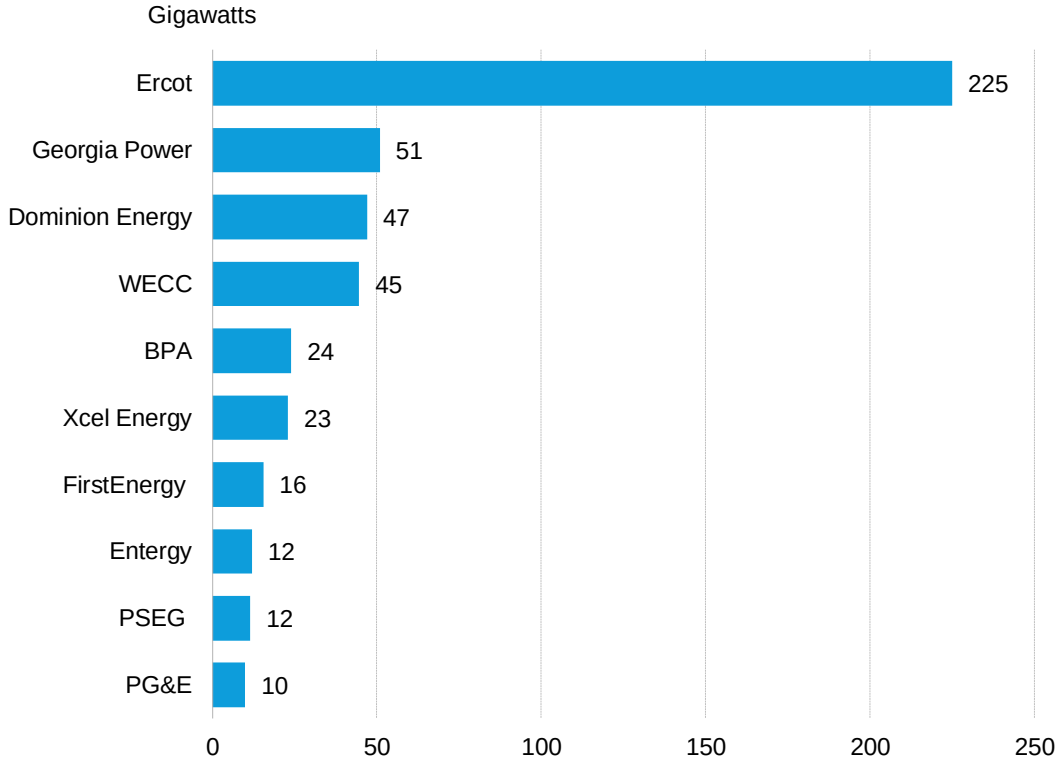
Data centers are returning electricity demand to structural growth

Drivers of electricity demand growth, Economic Transition Scenario



Source: BloombergNEF

Large load queues in selected US grids



Source: BloombergNEF, company filings, press releases

Speed to power is becoming a scarce resource for data centers

Data centers have yet to scale in demand response markets



Source: BloombergNEF, PJM, Monitoring Analytics.

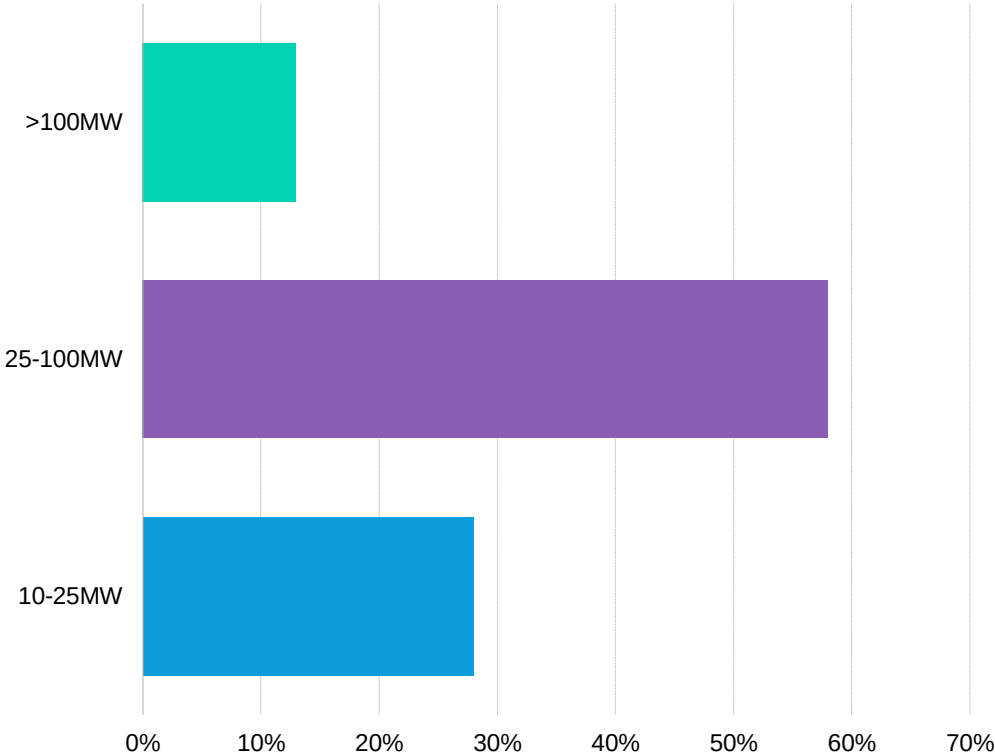
Data center flexibility is highest where workloads and infrastructure are controllable

		Workload type			Ownership type		
Category	Operational lever	AI training	AI inference	Traditional cloud services	Co-location	Self-build (hyperscaler)	Self-build (non-hyperscaler)
Shift	Schedule – run flexible jobs to lower-stress hours	High	Low	Medium	Low*	High	Medium
	Steer – route workloads to another facility or region	High	Medium	High	Low*	High	Low
	Soak – change when cooling power is used	Medium	Medium	Medium	Medium	High	Medium
Shape	Store – use batteries to change grid draw	High	High	High	Medium	High	Medium
	Supply – use dispatchable on-site or contracted supply	High	High	High	Medium	High	Medium
Shed	Shave – temporarily reduce non-critical facility or IT loads	Medium	Low	Medium	Medium	High	Medium
	Slow – reduce compute speed or throughput temporarily	Medium	Low	Low	Low*	Medium	Low
	Shed – curtail load during grid stress events	Medium	Low	Low	Low*	Medium	Low
Shimmy	Stabilize – ride through short disturbances and reconnect smoothly	High	High	High	High	High	High

Source: BloombergNEF, Lawrence Berkeley National Laboratory, Electric Power Research Institute (EPRI)

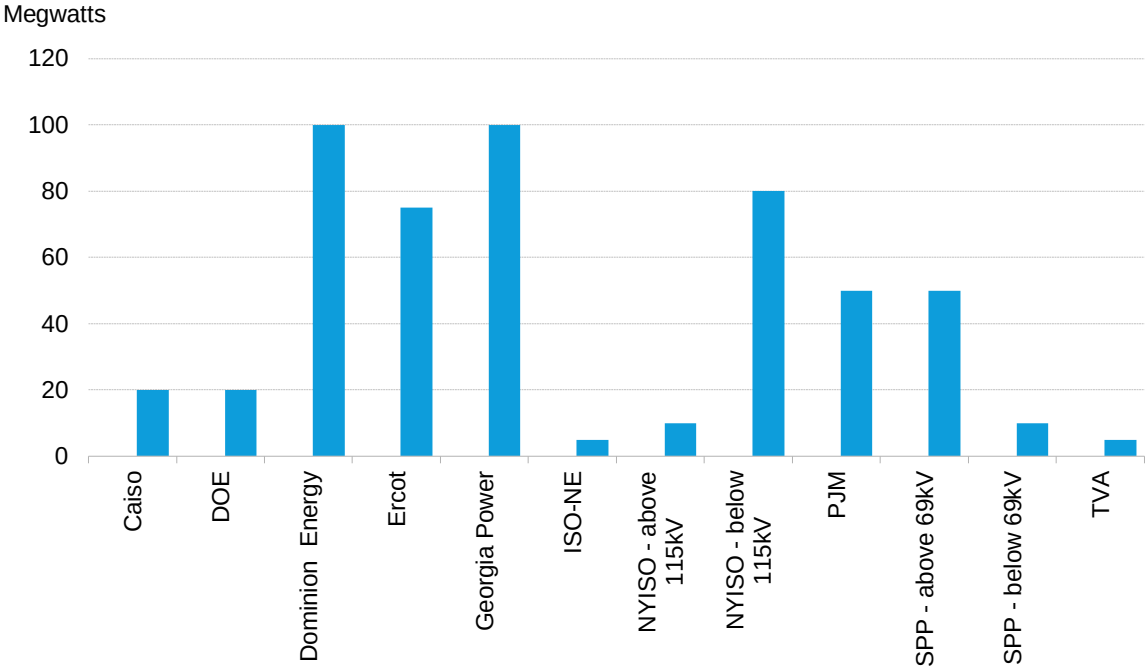
Grid connection rules must first define what counts as a large load

Results of NERC survey on the threshold for a large load



Source: National Electric Reliability Council (NERC)

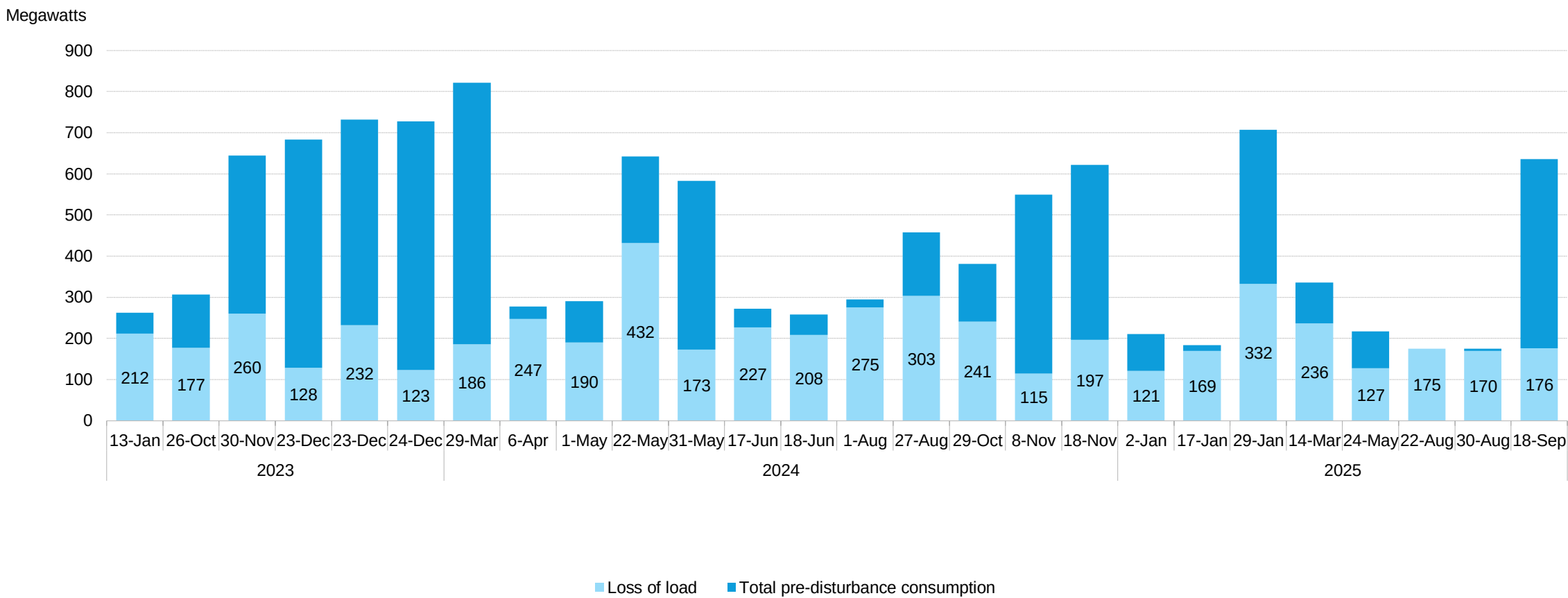
Varied large load thresholds from the US



Source: BloombergNEF, Utility reports and filings, Nation. Note: Caiso is California ISO, ISO is independent system operator, DOE is department of energy, Ercot is Electric Reliability Council of Texas, NYISO is New York ISO, ISO-NE is ISO New England, PJM is PJM interconnection, SPP is Southwest Power Pool, TVA is Tennessee Valley Authority, kV is kilovolt.

Sudden changes in large loads turn a capacity issue into a stability risk

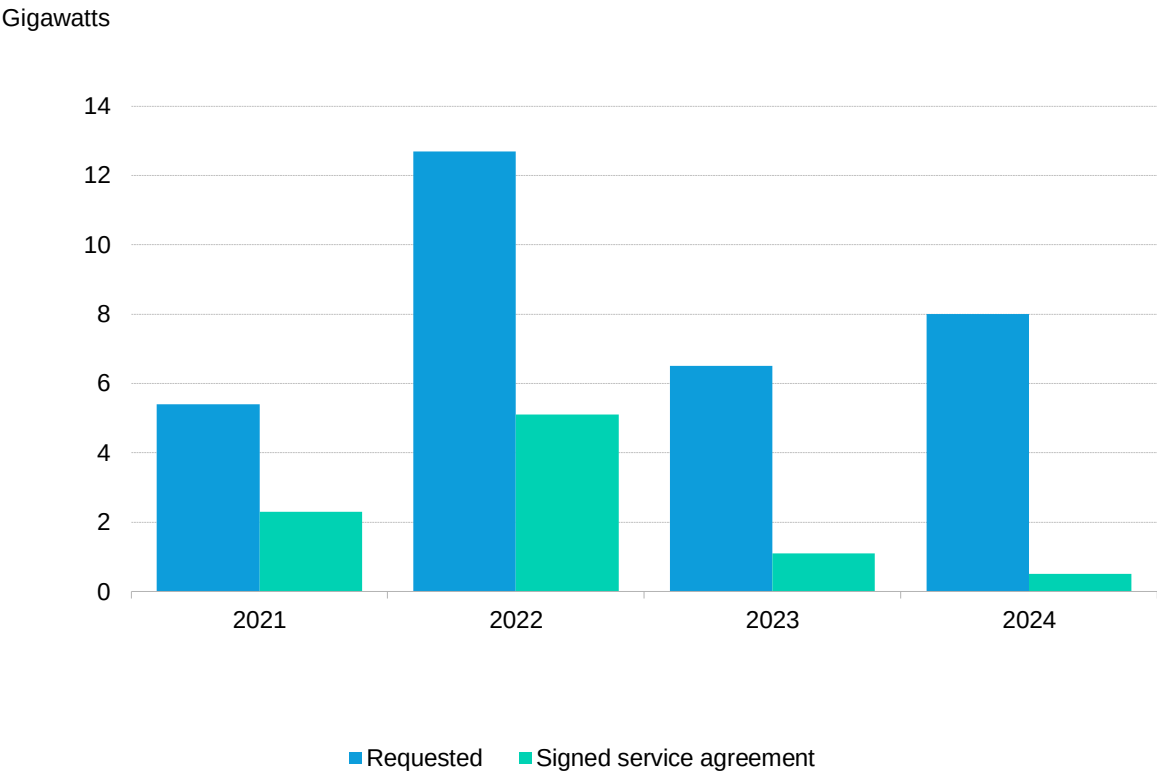
Ercot large load ride-through events between 2023-2025



Source: BloombergNEF, Electric Reliability Council of Texas (Ercot). Note: Total pre-disturbance consumption is approximated to the nearest 5 megawatts.

Sudden changes in large loads turn a capacity issue into a stability risk

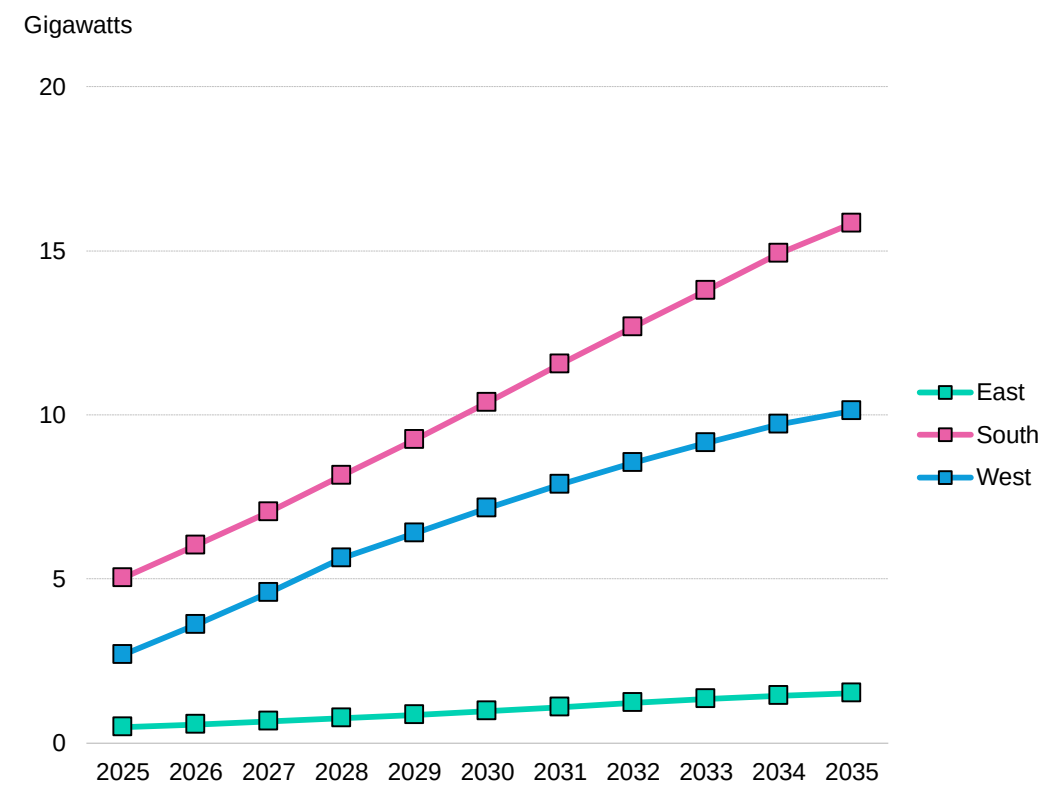
SPP connection queue for large loads



Source: BloombergNEF, National Electric Reliability Council, Southwest Power Pool. Note: Statistics incomplete for 2023 and 2024. Figure 12 title was updated March 13, 2026 to reflect that the data is for all large loads not just those less than 100MW.

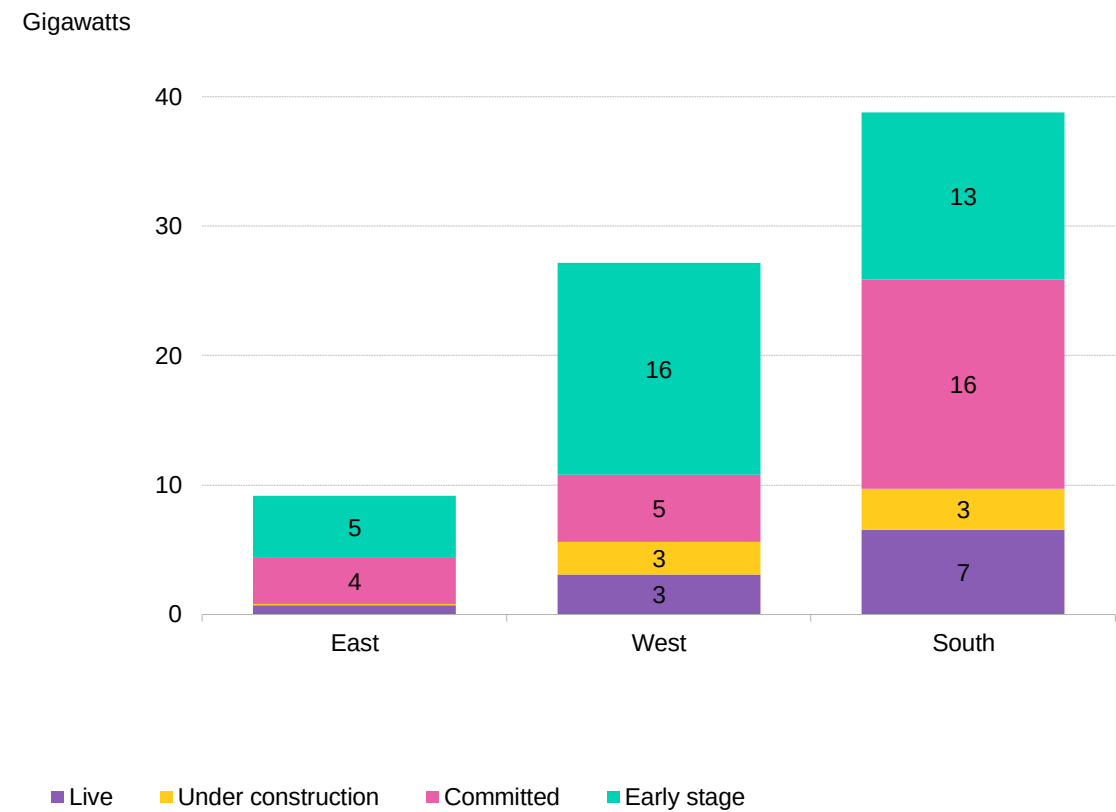
PJM's greatest risk is forecast error, not load growth alone

Average hourly data center load forecast by PJM subregion



Source: BloombergNEF, DCByte

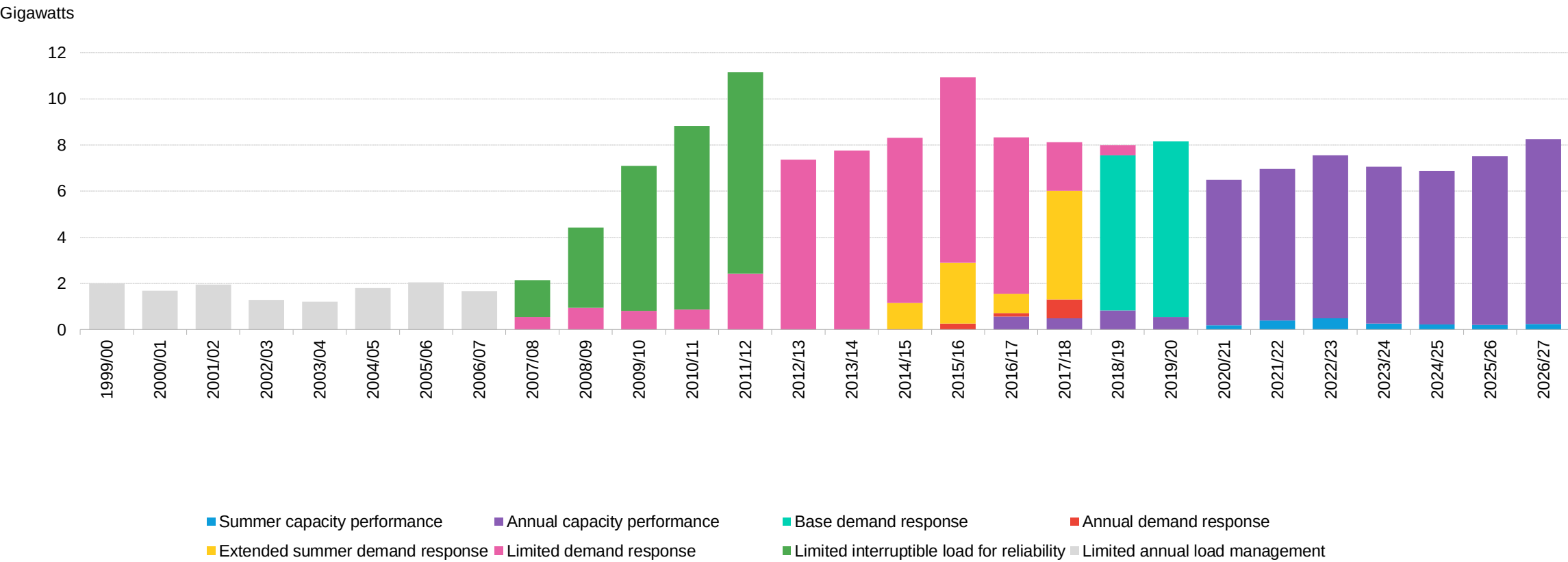
Data center pipeline by stage and subregion



Source: BloombergNEF, DCByte

Demand response can help, but data center participation remains low (1)

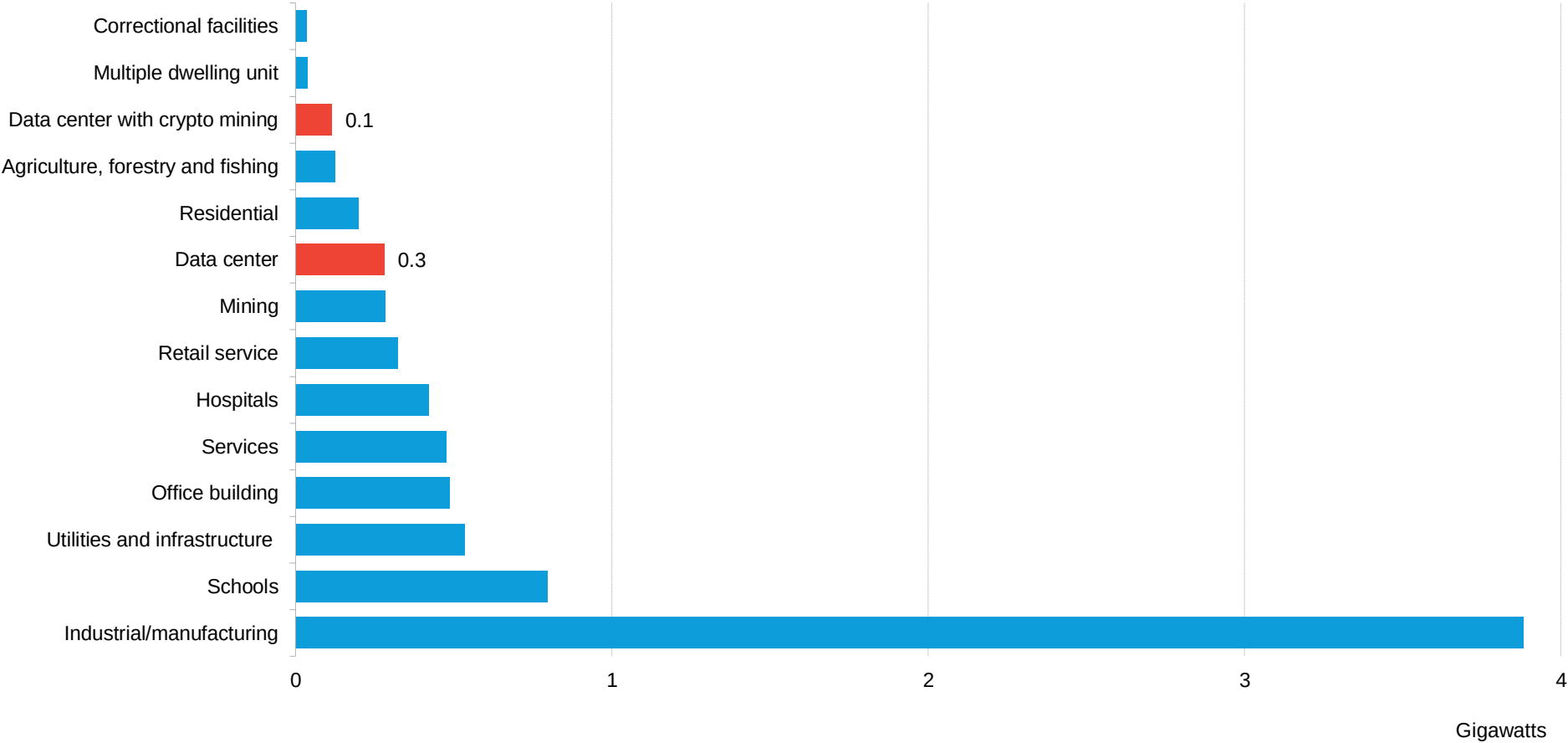
Demand response committed by delivery year



Source: BloombergNEF, PJM, Monitoring Analytics

Demand response can help, but data center participation remains low (2)

Nominated demand response installed capacity by business segment in the 2025/26 delivery year



Source: BloombergNEF, PJM, Monitoring Analytics

FOUR

HOW POWER-MARKET REFORM MUST RESPOND



Future power markets must price five distinct sources of value

BNEF assessment of power market and policy mechanisms

Each mechanism is scored out of 15 based on three criteria	Impact on investment signals	Wider system effects	Political/practical viability	Total
Low-carbon revenue-certainty policies				
Low-carbon tolling or similar measures				
Low-carbon power-purchase deals				
Low-carbon capacity payments				
Low-carbon mandates and certificates				
General capacity mechanisms				
Ancillary services				
Low-carbon subsidies				
Scarcity-based wholesale market				
Pay-as-bid wholesale market				
Separate wind and solar market				
Fossil-fuel subsidies				

Source: BloombergNEF



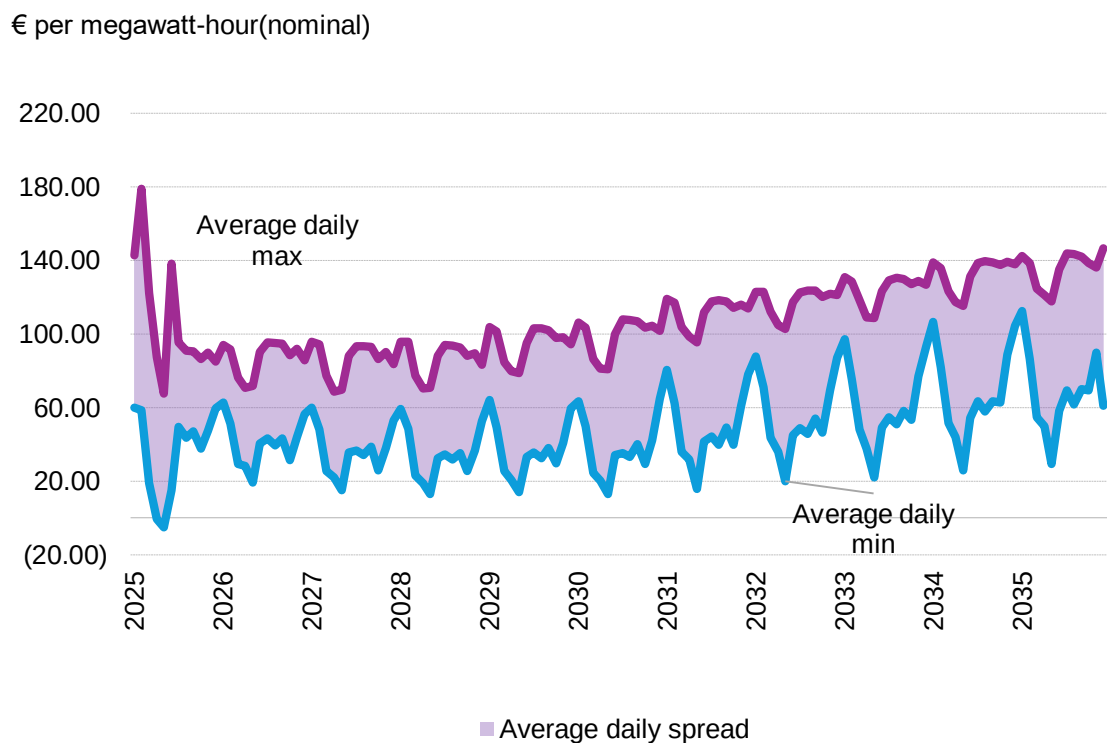
Policy and market mechanisms discussed in this report

Click on link below to jump to section	Energy	Reliability	Clean energy transition
	Sufficient electricity is produced to meet demand	It supplies electricity continuously under normal and emergency conditions	It undergoes shifts that markets alone may not deliver, in this case a move to a low-carbon technology mix
Scarcity pricing			
Pay-as-bid wholesale market			
Capital-cost subsidies for fossil fuels			
Ancillary services			
Capacity mechanisms			
Clean energy subsidies			
Separate wind and solar wholesale market			
Low-carbon capacity mechanisms			
Revenue stabilization mechanisms			
Power-purchase agreements			
Tolling and floor-plus-share agreements			
Mandates and certificates			

Source: BloombergNEF

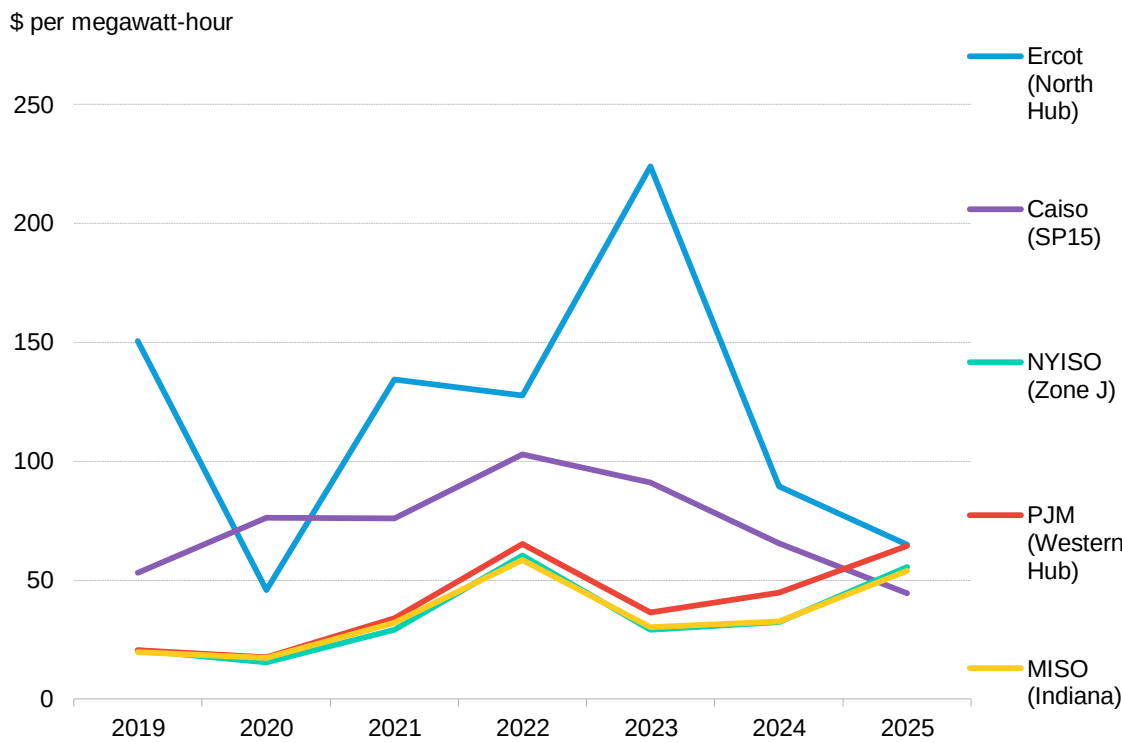
Spot prices guide dispatch; capacity mechanisms secure future availability

Spain – monthly power price spreads in BNEF’s Base scenario



Source: BloombergNEF. Note: 2025 is a weighted average of historical figures and modelled outputs. MWh is megawatt-hour.

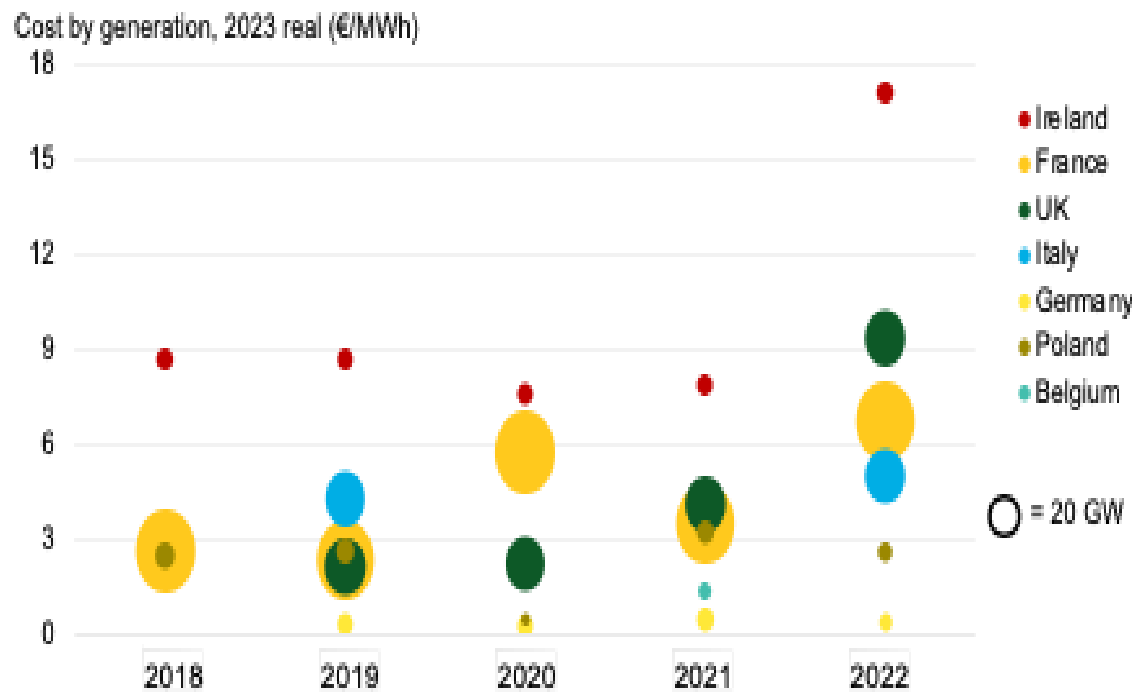
Annual price volatility (annual average of daily min-max spread)



Source: Innotap, BloombergNEF

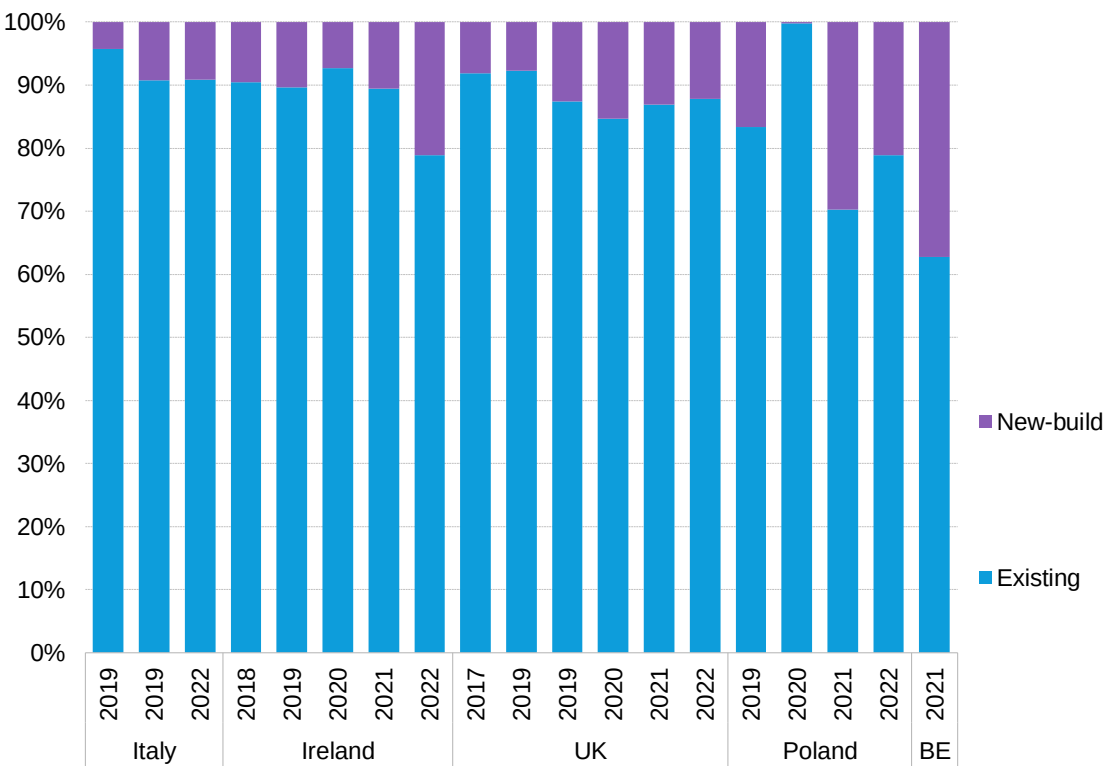
Capacity mechanisms are expanding, but their costs and designs vary widely

Value of capacity mechanism divided by 5-year average annual generation



Source: BloombergNEF

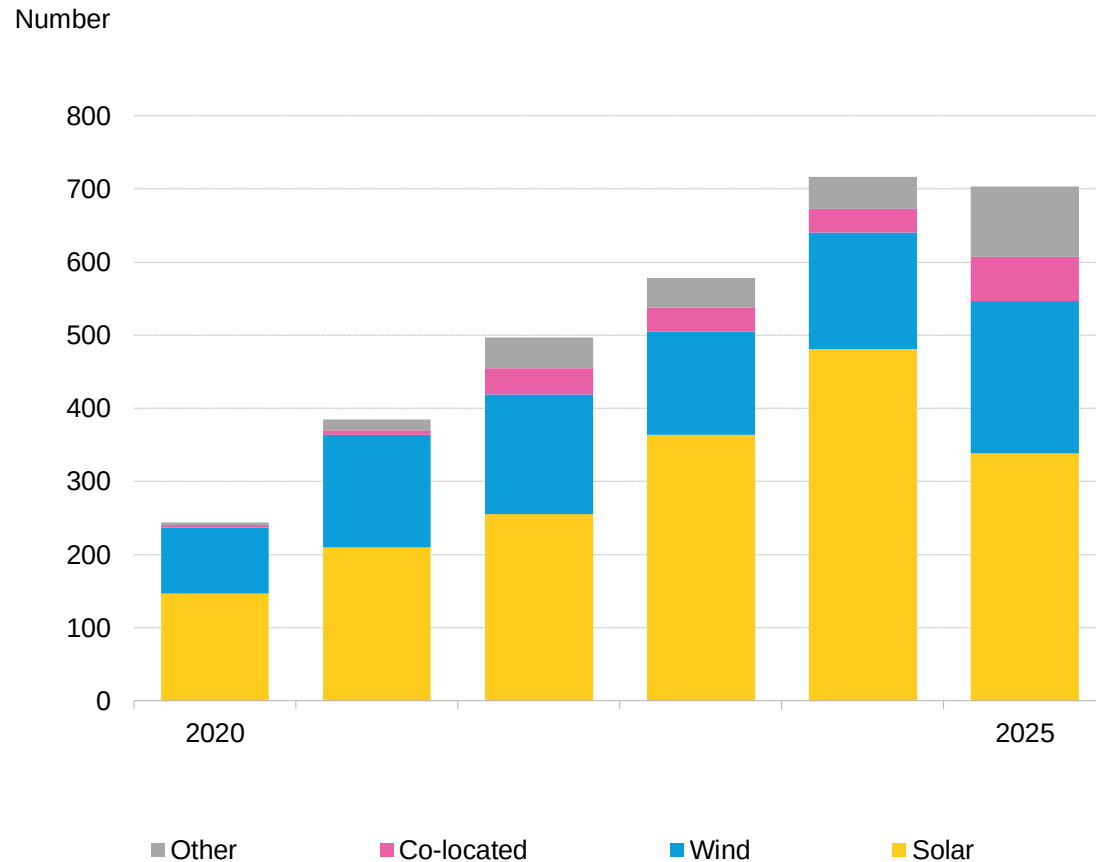
New vs existing contracted capacity by auction year



Source: BloombergNEF

Long-term contracts should reduce financing risk without muting market signals

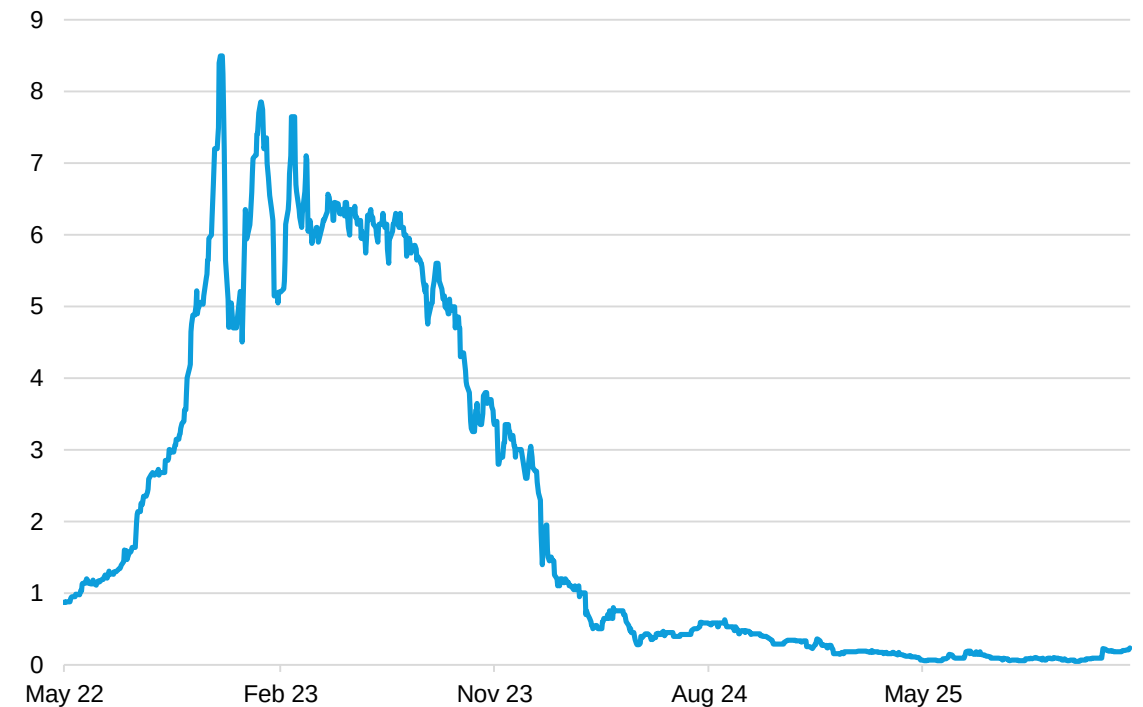
Number of annual corporate PPAs signed per year



Source: BloombergNEF

Guarantee of origin prices for renewables in Europe

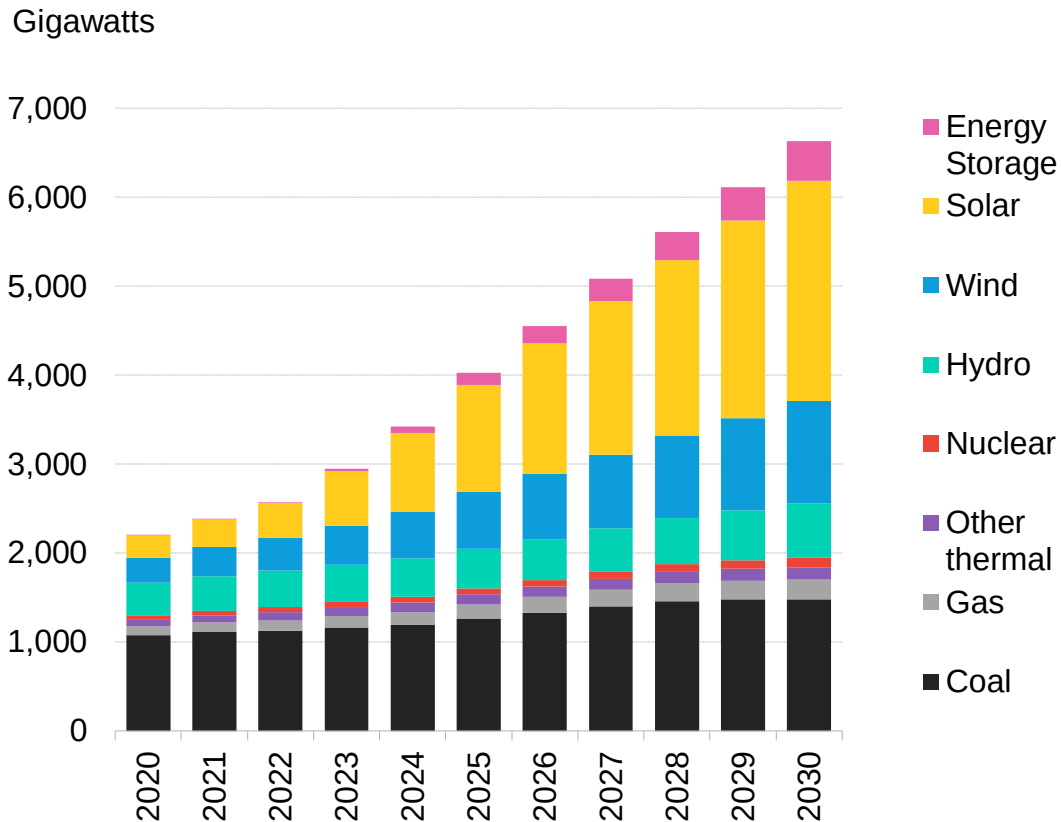
€ per megawatt-hour



Source: Argus Media Group, Bloomberg Terminal.

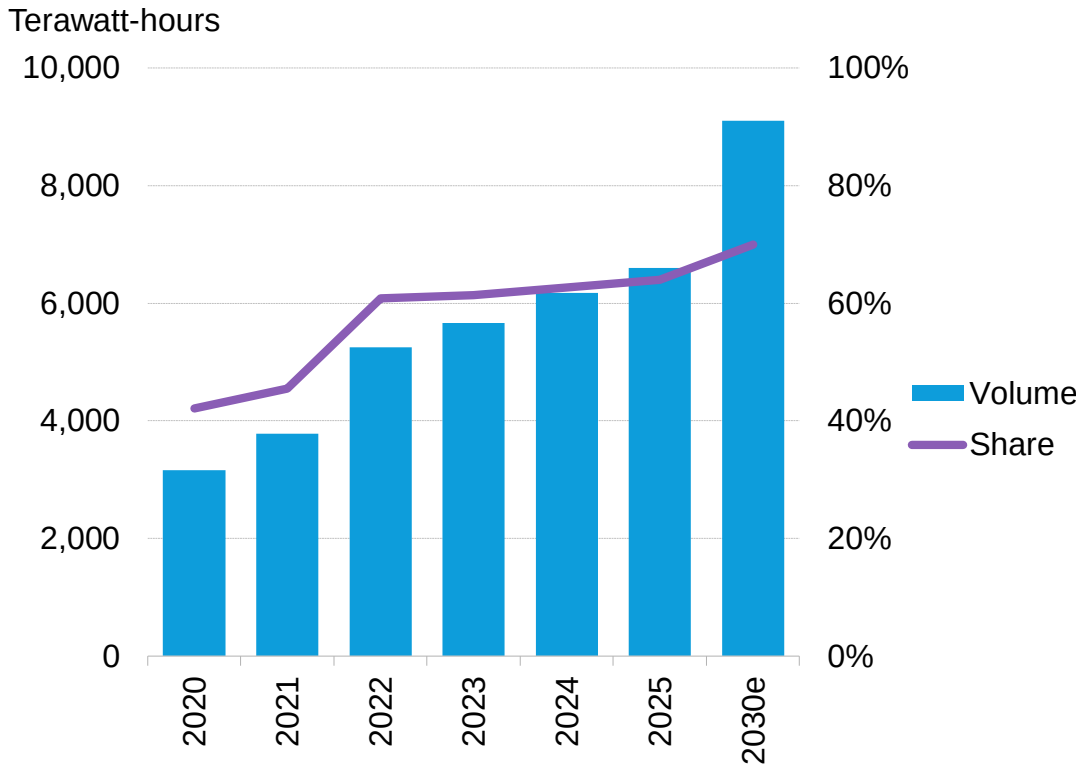
China is pursuing capacity expansion and market integration in parallel

China's cumulative installed capacity by technology



Source: BloombergNEF, China Electricity Council, National Energy Administration

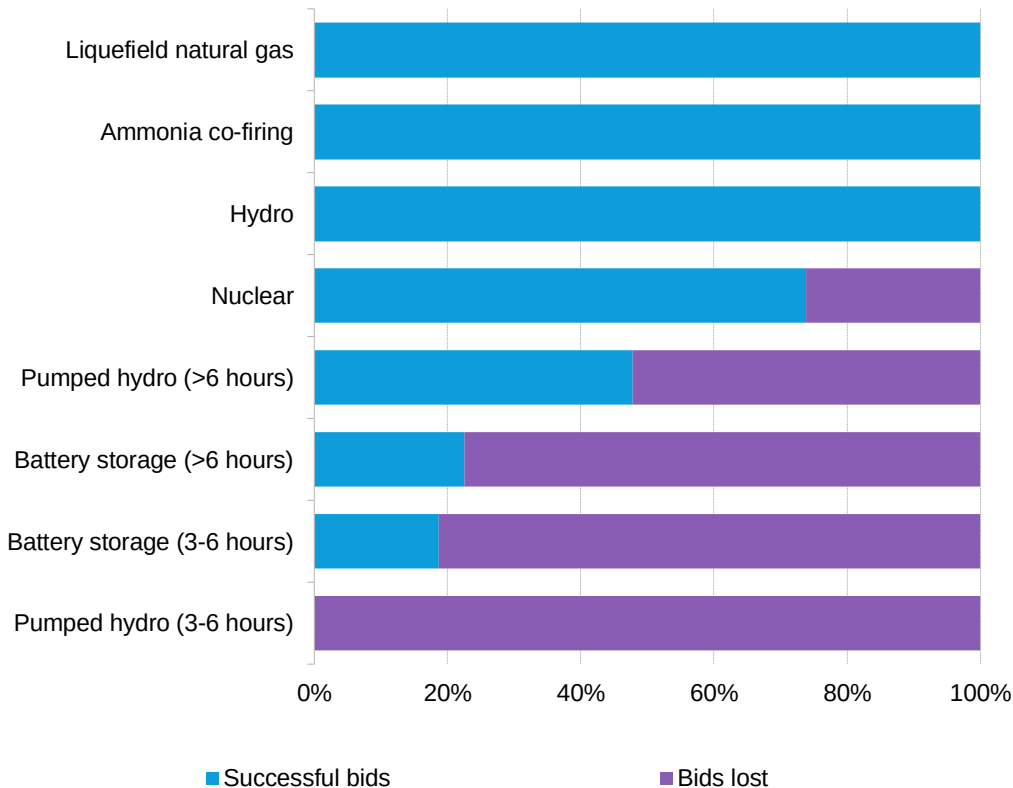
China's traded power volume and share of total power demand



Source: BloombergNEF, China Electricity Council. Note: Power demand is projected to reach around 13,000 terawatt-hours by 2030, according to estimates from the China Electricity Council.

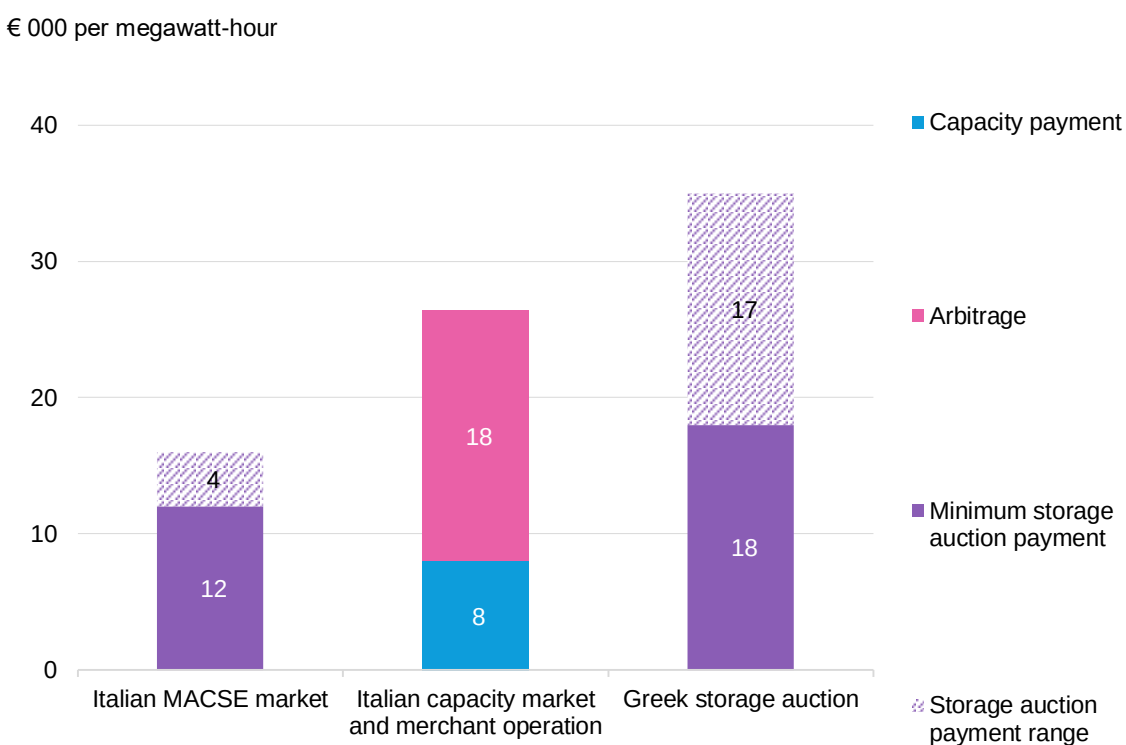
APEC economies need common reform principles, not a single market template

Breakdown of bid success rate in Japan's second long-term decarbonization auction, by technology



Source: BloombergNEF, Organization for Cross-regional Coordination of Transmission Operators (OCCTO)

Comparison of annual revenues for 4-hour duration batteries in Greece and Italy



Source: BloombergNEF, Raaey

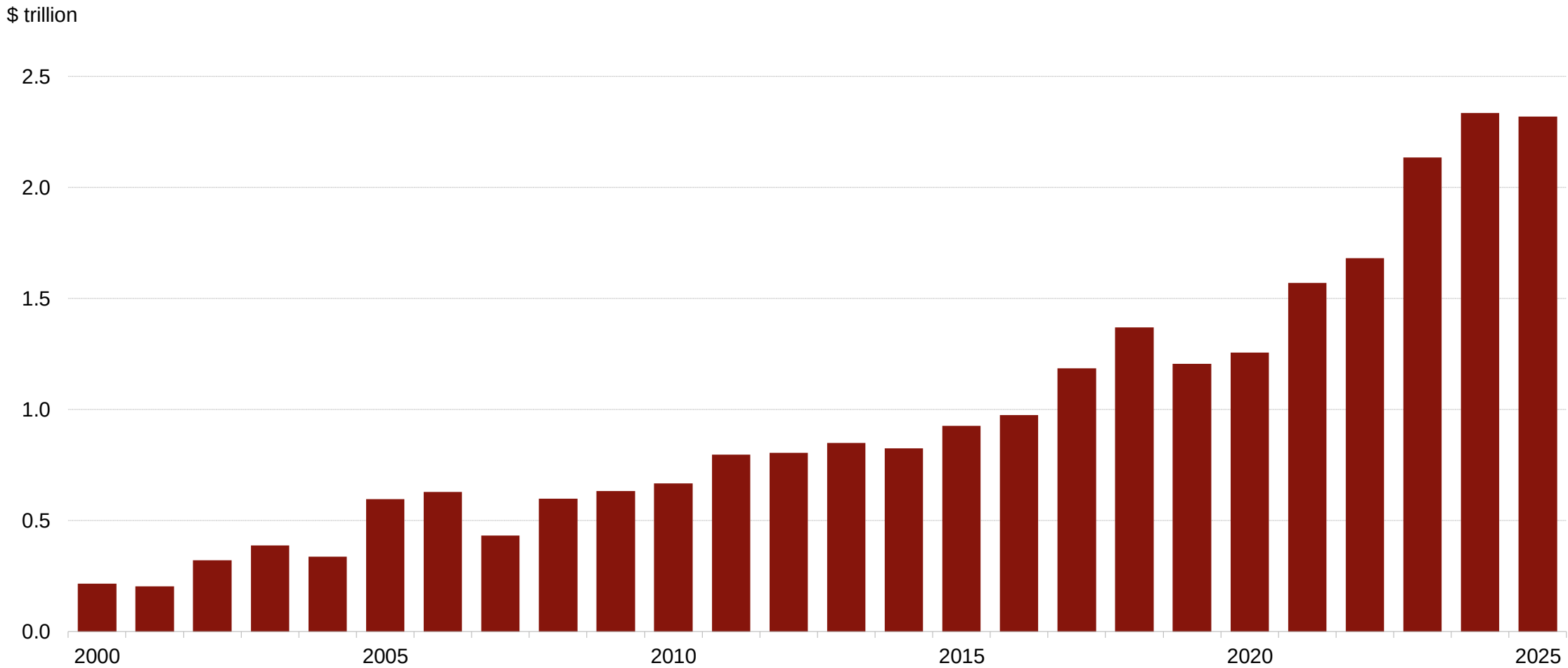
FIVE

CONCLUSION



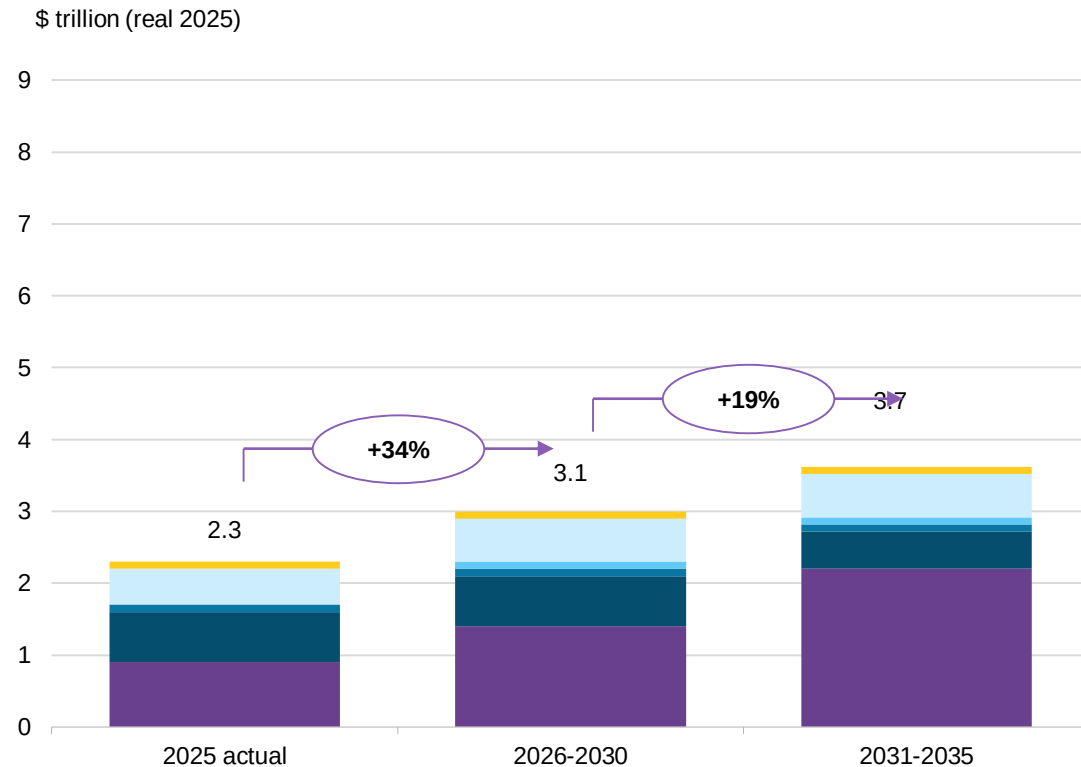
Climate physical impacts will only get worse

Annual global climate damages



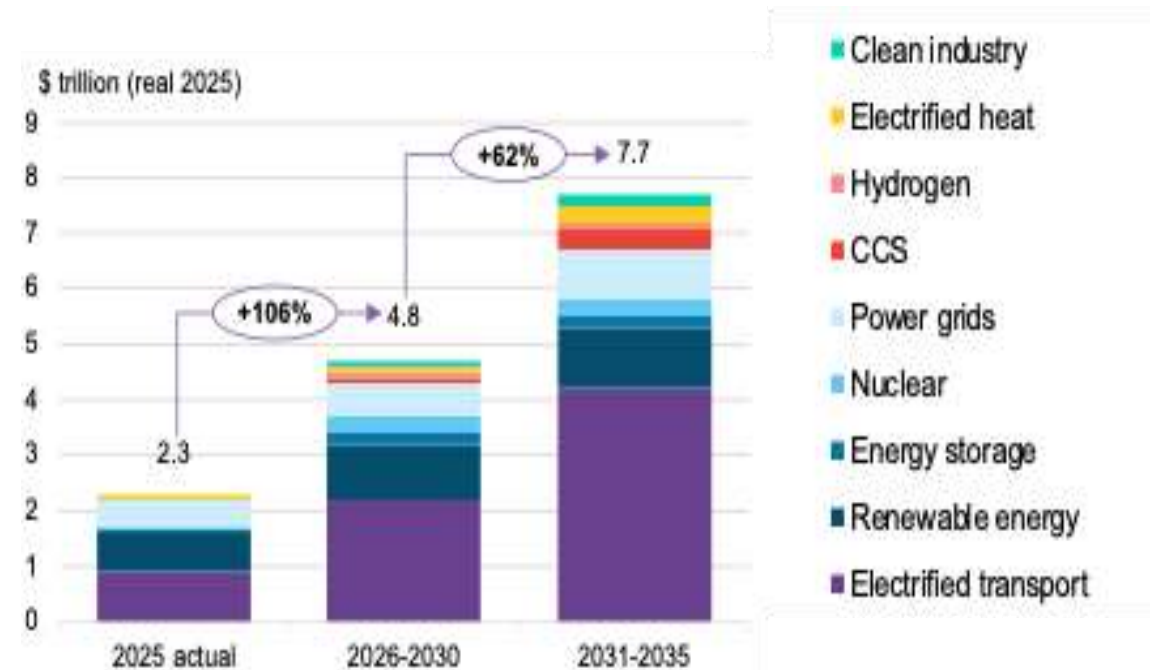
Competitive advantage will come from lower system costs, not just cheaper generation

Energy transition investment, 2025 actuals versus required annualized levels in NEO ETS scenario



Source: BloombergNEF

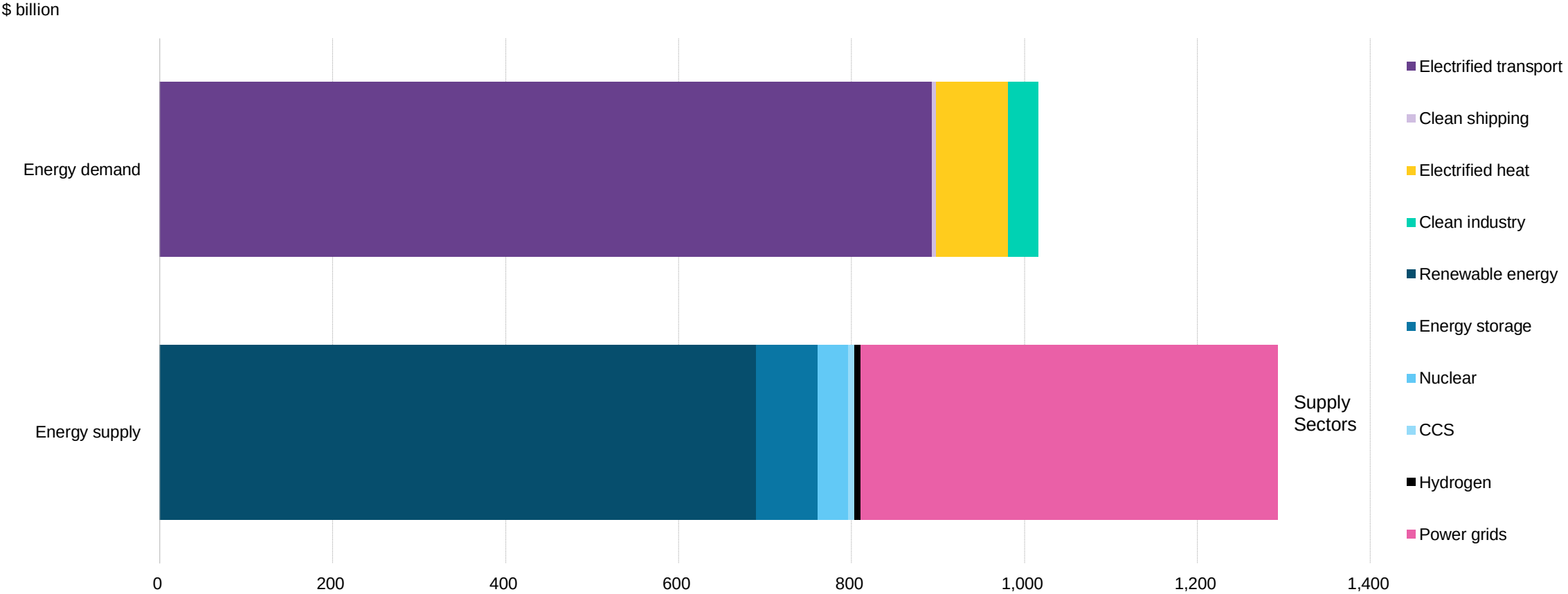
Energy transition investment, 2025 actuals versus required annualized levels in NEO NZS scenario



Source: BloombergNEF

Competitive advantage will come from lower system costs, not just cheaper generation

Investment comparison: supply-side versus demand-side in 2025



Source: BloombergNEF



And look at the horizon

You have to look at your feet

In times of uncertainty...

Optionality has value!

Image: Cheung 2016





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THANK' S

